A free-space convolution on a staggered mesh for the initial field of a uniformly moving particle bunch

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Abstract

Particle-in-cell simulations of laser wakefield acceleration need the initial electromagnetic field of a relativistic bunch before the first push, and the current workaround, a few field-free steps from a zero-field state, has no physical basis. We compute the present-time field of a uniformly moving bunch as a zero-padded free-space FFT convolution of a TSC charge deposit with the sampled TSC-convolved Liénard–Wiechert kernel, read out on the staggered Yee grid, and we measure its accuracy and cost. Against an independent quadrature reference the mesh reaches 3×10^{-4} relative error within eight cells of the bunch axis for a stationary particle and within forty-eight cells for $\gamma = 10$ perpendicular to the beam. The deposit dominates the compute time at large particle counts, linear in N, while the FFT scales with the padded grid; the total for 2×10^6 particles in a 20-cell box is 64.8 ms on one V100. An opt-in near shell replaces the mesh inside the TSC support with a closed-form evaluator, assembles exactly to roundoff, and is bitwise independent of the MPI rank count. The method writes a float32 openPMD checkpoint that restarts under a stock PIConGPU.

1 Introduction

When a particle-in-cell (PIC) simulation of laser wakefield acceleration starts, the driven bunch already moves relativistically through the plasma [Esarey et al., 2009], so the first push must act on the correct electromagnetic force. In practice the start state carries no field: simulation frameworks either start field-free and let the wake build up in the first few steps, or compute the initial field by evaluating the closed-form expression of each source charge at every observation point. The first option inserts a transient that the physics of a uniformly moving bunch does not contain, and the second is too slow once the staggered mesh holds thousands of observation points per source.

The present-time field of a uniformly moving charge is known in closed form [Jackson, 2007, Landau and Lifshitz, 1975], and the field of a bunch of N such charges is the linear sum over sources. Evaluated on a mesh that also carries a TSC deposit [Birdsall and Langdon, 1985, Hockney and Eastwood, 1981], the sum becomes a discrete convolution between the deposited charge and the TSC-convolved point-charge kernel. A zero-padded free-space FFT evaluates that convolution in $\mathrm{O}(M^3 \log M)$ transforms with M the padded grid size, independent of N in the transform itself. `picBunch` implements exactly this operation, with staggered Yee read-out, an opt-in near shell that evaluates the in-support integral in closed form, an MPI-parallel component- and tile-parallel design, and a checkpoint writer that restarts under a stock PIConGPU [Burau et al., 2010, Huebl et al., 2015].

This paper gives the discrete recipe, its physical consistency checks, and measured accuracy and cost. Section 2 defines the field and Section 3 the algorithm. Section 4 checks the kernel against the Liénard–Wiechert solution, Lorentz contraction, Gauss's law, the Yee staggering, and the $B = \boldsymbol{v} \times \boldsymbol{E}/c^2$ limit. Section 5 separates the error into mesh/deposit, velocity binning, near-source quadrature, far-tail, and output-precision terms, and states the domain of validity of each near-shell mode. Section 7 measures cost and MPI scaling. Appendices give the full derivations, the secondary validation sweeps, the large-domain slices, and the multi-velocity measurements.

2 Physical model and field definition

The computed quantity is a single linear integral. For a bunch of source charges at positions $\mathbf{x}_p$ with signed charge q_p and a shared dimensionless velocity $\mathbf{u} = \mathbf{p}/(mc)$, the field at a staggered observation

point $\mathbf{x}_{\text{obs}}$ is the TSC-convolved uniformly-moving point-charge field

$$F_c(\mathbf{x}_{\text{obs}}) = \sum_p q_p \int_{[-1.5,1.5]^3} W(\mathbf{s})\, K_{\text{point},c}\big(\mathbf{x}_{\text{obs}} - \mathbf{x}_p - \mathbf{s}\,\mathbf{cell},\ \mathbf{u}\big)\, d\mathbf{s}^3, \tag{1}$$

where $W(\mathbf{s}) = W_x(s_x)W_y(s_y)W_z(s_z)$ is the separable TSC shape (Section 2.3) and K_{point} is the per-unit-charge field of one uniformly moving charge. The sum over particles is exact because the integral is linear in the source charge; the algorithm of Section 3 is a fast evaluation of this one integral.

2.1 The moving point kernel

The kernel is the present-time Liénard–Wiechert field of a uniformly moving charge in the present-position form [Jackson, 2007, Landau and Lifshitz, 1975]. In the lab frame, with $\mathbf{R} = \mathbf{x}_{\text{obs}} - \mathbf{x}_{\text{src}}$,

$$\gamma = \sqrt{1 + |\mathbf{u}|^2}, \qquad \boldsymbol{\beta} = \mathbf{u}/\gamma, \qquad \text{invg}^2 = 1/(1 + |\mathbf{u}|^2) = 1 - \beta^2, \tag{2}$$

$$D = |\mathbf{R}|^2\, \text{invg}^2 + (\mathbf{R}\cdot\boldsymbol{\beta})^2, \tag{3}$$

$$\mathbf{E}_n = \text{invg}^2\, \mathbf{R}/D^{3/2}, \qquad \mathbf{B}_n = \boldsymbol{\beta} \times \mathbf{E}_n. \tag{4}$$

For $\mathbf{u} = \mathbf{0}$ exactly, $\mathbf{E}_n = \mathbf{R}/|\mathbf{R}|^3$ and $\mathbf{B}_n = \mathbf{0}$; no ε is added to any denominator. The kernel is homogeneous of degree -2 in $\mathbf{R}$. A bunch with a velocity spread needs one convolution per velocity class (Section D.1, Appendix D); the validated scope is a single shared $\mathbf{u}$.

2.2 The unit chain

The kernel normalizes the per-unit-charge field as $E_n = 4\pi\varepsilon_0\, E/q$ and $B_n = 4\pi\varepsilon_0 c\, B/q$, so $B_n = \boldsymbol{\beta} \times E_n$. The mesh produces the per-unit-charge field with $\mathbf{R}$ in metres, already summed over the deposited charge q_p in coulombs. The chain to physical SI is

$$E_{\text{SI}}[c] = F_c[c]\, h^3/(4\pi\varepsilon_0), \quad B_{\text{SI}}[c] = F_c[c]\, h^3/(4\pi\varepsilon_0 c), \tag{5}$$

with $\varepsilon_0 = 8.8541878128 \times 10^{-12}\,\text{F/m}$ and $c = 299792458\,\text{m/s}$. The h^3 factor is the Riemann volume of the sampled continuum integral (Section 3).

2.3 The TSC shape

W is the quadratic B-spline, with argument in cell units, support $[-1.5, 1.5]$ per axis, and unit integral per axis:

$$W(x) = \begin{cases} 3/4 - x^2 & |x| < 1/2, \\ \frac{1}{2}(3/2 - |x|)^2 & 1/2 \le |x| < 3/2, \\ 0 & \text{otherwise.} \end{cases} \tag{6}$$

3 Numerical algorithm

The mesh evaluates Eq. (1) by a discrete deposit, a sampled kernel, a free-space FFT convolution, and a staggered read-out. Let N be the bunch-box nodes per axis, $M = \text{pad} \cdot N$ the padded FFT grid per axis, h the node spacing in cell units (node i sits at $ih\,\mathbf{cell}$), and $\mathbf{cell} = (dx, dy, dz)$. All precision-critical work is in double precision.

3.1 Deposit

Each source charge is deposited onto the zero-padded M^3 grid with the TSC shape, at the cell-unit spline argument $W((\mathbf{x}_i - \mathbf{x}_p)/\mathbf{cell})$, which is correct for any h. The result $\rho[i] = \sum_p q_p\, W(ih - \mathbf{p}_{\text{cell}})$ lives in the N^3 sub-box of the padded grid, indexed with M-stride; the rest is zero. A fixed reduction order is required for MPI-layout invariance; the single-rank path uses `atomicAdd`.

3.2 Kernel sampling

The six TSC-convolved kernels are sampled on the same padded M^3 grid, one thread per FFT point, with the centered (negative-wrapped) index $d_a = a$ for $a < M/2$ else $a - M$. For each of the six components c, the kernel is evaluated at the physical separation

$$\mathbf{R} = (d_a\, h + \delta_{c,a})\, \mathbf{cell}, \tag{7}$$

where δ_c is the Yee staggered offset (Section 3.4). The kernel is therefore sampled at the same node spacing h as the deposit, with the shape one grid cell of its own axis wide. Non-finite samples are set to zero to avoid $0 \times \infty$ in the FFT.

3.3 FFT convolution and normalization

The convolution is a free-space, zero-padded FFT convolution, with the discrete convolution theorem in the unnormalized form of the vendor documentation [Press et al., 1986, NVIDIA Corporation, 2024]. The plan is built once, unnormalized, and reused for all thirteen transforms: one forward for ρ, six forwards for the kernels, and six inverse transforms. The unnormalized round trip is scaled by M^3; the read-out divides by $h^3 M^3$, because the sampled convolution evaluates the continuum integral of Eq. (1) by a Riemann sum whose nodes are spaced h cells apart, so each sample covers a physical volume h^3 (in cell units).

Padding. The grid is zero-padded to pad ≥ 2 so that the cyclic convolution does not wrap [Press et al., 1986]. With pad $= 1$ the cyclic convolution wraps and a 9.275×10^{-3} periodic-image artefact appears for a single stationary particle in a 20^3 box; at pad ≥ 2 the error is at machine precision ($\leq 10^{-15}$). The artefact grows for beamed sources: at pad $= 1$ the 100-source $\gamma = 10$ cloud shows a 7.5×10^{-1} worst-component error (Appendix B).

DC and Nyquist. For the R2C/C2R half-spectra the DC bin and the seven self-conjugate even Nyquist corners must be exactly real; the intermediate Nyquist planes are Hermitian-paired, not real. The R2C/C2R path matches the C2C path to 1.665×10^{-16} over all six components (Appendix B).

3.4 Read-out

The six field components live at the staggered Yee positions, in cell units:

$$\begin{array}{llllll} E_x & (0.5, 0, 0) & E_y & (0, 0.5, 0) & E_z & (0, 0, 0.5) \\ B_x & (0, 0.5, 0.5) & B_y & (0.5, 0, 0.5) & B_z & (0.5, 0.5, 0). \end{array} \tag{8}$$

The kernel is sampled at these offsets, so the convolution is already evaluated on the Yee grid. The read-out compacts the M^3 FFT result to the N^3 bunch box and applies the h^3/M^3 scale and the SI factor (Section 2). No further interpolation is needed.

3.5 Scale invariance

The kernel is homogeneous of degree -2, the deposit shape width equals the node spacing, and the Yee offsets are in cell units, so every relative metric depends only on r/h, the distance in cell units. At a fixed distance in cells the error is independent of h: refining h with the distance held in cells changes nothing, and the measured relative errors at $h = 1,\ 1/2,\ 1/4,\ 1/8$ are identical to machine precision for $\gamma \in \{1, 10, 100\}$, aspect 1 and 4. At a fixed physical distance r, refining h increases r/h and does reduce the error, at the cost of M^3 FFT work in $1/h$. The controlling length for a beamed source is the beaming cone width d/γ: where that width is sub-cell, the error is set by the cone, not by the mesh (Section 5.5).

3.6 The near shell

The default output is the whole mesh. An opt-in near shell replaces the mesh inside a radius R (in mesh nodes of the compute grid) around each source:

$$F = F_{\text{mesh}}(\text{all}) - F_{\text{mesh}}(\text{near}) + F_{\text{near}}(\text{near}), \tag{9}$$

where the subtraction uses a per-tile local zero-padded FFT of the near sources (padding 2) and the near term is evaluated per (source, node, component) by one of two backends. `exact` is an adaptive FP64 TSC-convolution quadrature shared with the independent reference driver; `fast` is the closed-form evaluator below. The default is `none`; the envelope the radius must cover depends on h (Appendix C).

The fast evaluator splits the in-support integral of Eq. (1) into a singular part and a smooth remainder. The singular part is the eight-term asinh box of the kernel's divergence, O(1) per pair, built from the double integral

$$T(A,B,u) = \int_0^A \int_0^B \frac{\mathrm{d}X\,\mathrm{d}Y}{\sqrt{u^2+X^2+Y^2}} = \begin{cases} A\,\mathrm{asinh}(B/A) + B\,\mathrm{asinh}(A/B), & |u| < 10^{-300}, \\ A\,\mathrm{asinh}\big(B/\sqrt{u^2+A^2}\big) + B\,\mathrm{asinh}\big(A/\sqrt{u^2+B^2}\big) & \\ \quad -u\,\arctan\big(AB/(u\sqrt{A^2+u^2+B^2})\big), & \text{otherwise.} \end{cases} \tag{10}$$

with $A, B > 0$; each box face has 8 corners and 24 such calls per component, so the eight-term box is O(1) per pair. The form is even in u, so the $|u| \to 0$ branch must fire on $|u|$, not u (a negative u is a field point beyond a box face). The remainder is a plain tensor Gauss–Legendre quadrature on the subtracted integrand $(S - W^*)K$ over the TSC knot panels, each axis split at the TSC knots and at the anchor point s^*; the default order is 5 (8000 nodes per in-support pair). For a moving bunch the box is γ-stretched for the axial direction, and the oblique direction uses a rest-frame octant quadrature of the singular box at order 8 (4096 nodes) plus the same remainder. Host and device use the same algebra; the full derivation is in Appendix A.

Assembly. Single-source tests at a tile centre and a tile boundary give an assembly residual of $2{\times}10^{-12}$ to $1.5{\times}10^{-11}$ SI ($4{\times}10^{-16}$ relative), flat with distance. A local FFT of the near set against the global FFT agrees at 10^{-15} for margins 0, 2, 4 and a 100-particle cloud. A direct reproduction with 100 particles at radius 3 gives an assembled field exact to 4.6×10^{-13} over the whole grid, while the plain mesh on the same plane is wrong by 5.6×10^{-1}. The fast backend changes no far number: the mesh-only field from the fast and the exact backends is bitwise identical, and the total field outside the near tiles is bitwise equal, so the tile subtraction stays intact. The seam jump excess at the fast/exact crossing, on a 16^3 grid, $R = 3$ cells, tile 4, cells 10^{-7} m, is $2.03{\times}10^{-10}$ per component for a stationary single particle ($2.8{\times}10^{-10}$ for the 64-source cloud).

3.7 The far-tail margin

For a global domain that extends beyond the compute box, the zero fill must not start until the field is below tolerance. The margin is placed at the smallest centre distance d where

$$E_{\mathrm{tail}}(d) = \gamma_{\max}\, k \left(\frac{|Q|}{d^2} + \frac{2|\mathbf{p}|}{d^3} + \frac{3.5\,\|D\|_2}{d^4} \right) \le \text{tolerance} \cdot E_{\mathrm{peak}}, \tag{11}$$

with Q the total charge, $\mathbf{p}$ the dipole, $D_{ij} = \sum q\,(3x_i x_j - |\mathbf{x}|^2\delta_{ij})$ the traceless quadrupole from the actual particle positions, $\|D\|_2$ its largest absolute eigenvalue, and $\gamma_{\max} = \max\sqrt{1+|\mathbf{u}|^2}$. Beyond a measured switch radius the writer evaluates an asymptotic moving multipole rather than a static one: the exact Liénard monopole of the total charge moving at the charge-weighted mean velocity, plus the rest-frame dipole and quadrupole evaluated at the Lorentz-stretched observation point and boosted back to the lab frame. For $\beta = 0$ this reduces exactly to the static multipole and for one particle it reproduces the point kernel in every direction; for an extended bunch it is the leading moving multipole, not a full Liénard multipole sum. The output file therefore has no interior zero fill. The margin is reported at run time as a bound in V/m; the switch error against an independent direct charge sum is in Appendix D.1, and the moving point-kernel check is `Margin.MovingFarTailMatchesPointKernel`.

Spatial far-tail multipoles. The tail can use one multipole per spatial supercell instead of a single global expansion. The bunch is split into boxes of N PIC cells (`-multipole-group-cells`, default 16; 0 keeps one global multipole), one set of moments is accumulated per occupied box about its own centre, and the tail sums the group expansions. For a two-cluster $\gamma = 10$ bunch the worst far-tail error against the direct Liénard sum at $6R$, $8R$ and $12R$ falls from 1.72×10^{-3} (global) to 1.81×10^{-4} (4-cell groups). The evaluation cost grows with the number of occupied groups per tail node, so the group size controls

the accuracy-cost trade-off; large groups are cheaper, small groups are more accurate near the exact-box transition.

3.8 MPI parallelism

The program writes a global output domain larger than the compute box and computes the field in parallel across ranks. The written extent is set by `-global-grid` and `-global-origin`. The global array is never allocated: each rank owns a contiguous z-slab split into tiles of about 128 MiB, zero-filled, then filled with the compute-box overlap, the rest exact 0.0. All ranks open one openPMD series with `MPI_COMM_WORLD` and write disjoint `storeChunk` regions with collective flushes, so the output is a single dataset. A `-deterministic-deposit` flag gives a serial host deposit for bitwise comparisons.

The compute is parallel in two ways, neither of which is a domain-decomposed FFT. The mesh runs six independent component convolutions: every rank deposits and transforms the same ρ spectrum (replicated), rank r evaluates the components c with $c \bmod \text{size} = r$, and one `MPI_Allreduce` sums the six arrays. The near shell partitions the output tiles by linear tile index; each rank builds its own near-source list and local subtraction FFT for its tiles and writes only those corrections, and one `MPI_Allreduce` sums the correction arrays. The bunch is read once on rank 0 and broadcast, not redistributed. Because the mesh components are disjoint and each near correction element is owned by exactly one rank, the all-reduce order cannot change a value: the outputs are bitwise identical at 1, 2 and 4 ranks (Section 7.1).

4 Physical consistency

Five checks constrain the implementation against the physics it is built to reproduce. Each check names its measured value.

Liénard–Wiechert kernel. The kernel of Section 2 is the present-time Liénard–Wiechert field of a uniformly moving charge [Jackson, 2007]. The mesh reproduces the TSC-convolved Coulomb field of a single stationary particle to a worst relative error of 4.88×10^{-14} (2.28×10^{-25} absolute) at $d = 8, 16, 24, 32$ along all six axis directions, against the direct TSC stencil sum of the analytic Coulomb kernel (Coulomb anchor, Section 7).

Lorentz contraction. The field of a uniformly moving charge is compressed into a pancake of width d/γ perpendicular to the motion. The seven-velocity measurement (Appendix B, Fig. 9) tracks the pancake: at $\gamma = 10$ the mesh resolves the contracted field to 1.2×10^{-1} worst relative error on the mid-z plane of a 128^3 grid with 5×10^{-8} m cells, and the error grows with γ because the pancake becomes thinner than the mesh spacing (1.43 and 3.21 at $\gamma = 50$ and 100), not because of a solver error. Raising the resolution from 64^3 to 128^3 at the same physical domain leaves the relative errors unchanged.

Gauss's law. The discrete Gauss-law indicator, the Yee backward-difference div E residual normalised to the peak deposited ρ/ε_0, is 0.3534 for the mesh alone on the $N = 1000$ slice geometry ($\gamma = 1$) and 5.026 for $\gamma = 10$, against reference values 0.0904 and 0.0814. With the near shell enabled it moves to 0.0895 for $\gamma = 1$, reaching the reference value, and to 0.399 for $\gamma = 10$, a 12.6× drop that still leaves it 4.9× above the reference because the shell radius covers a fraction of the TSC support (Appendix C).

Yee staggering. The six field components are evaluated at the staggered positions of Section 3.4; the kernel sampling uses the same offsets, so the convolution is already on the Yee grid and the read-out applies no interpolation. The cross-validation against the PIConGPU `referenceOnly` field uses identical Yee offsets, grid, and origin, and the reference staggered point maps to a picBunch node by index with zero snap deviation (Section 6).

The $B = \boldsymbol{v}\times\boldsymbol{E}/c^2$ limit. The kernel satisfies $\mathbf{B}_n = \boldsymbol{\beta}\times\mathbf{E}_n$ exactly, and the mesh inherits it component by component. For a stationary bunch ($\gamma = 1$) every measured B component is exactly zero (Tables 8 and 18); for motion along $+y$ the component B_y is identically zero in the measured fields ($B_y \equiv 0$ in Figs. 13 and 9). The magnetic field at $\gamma = 10$ is about 10^{-10} of the E-dominated local field in the floored metric, which is why the floored error metric is electric-only there (Section 5).

5 Validation and accuracy

Every error figure names its reference. The three references stay apart throughout: the mesh M, the TSC-convolved continuum T (Eq. (1), the quantity the mesh is built to compute), and the point field P (the un-smoothed Liénard–Wiechert field). M against T is the discrete deposit and convolution error. T against P is the TSC smoothing, an $O(1)$ effect at $\gamma h \gg 1$. M against P is the total.

5.1 The references and the metrics

The arbiter is an independent TSC-convolved continuum, a tensor Gauss–Legendre quadrature with a TSC-knot split and a singularity-cancelling power map; its quadrature error is below the 3×10^{-4} gate in every configuration measured here. Two more references constrain the method externally: an order-12 independent continuum (refine 3) against which the CLI is measured, and the PIConGPU `referenceOnly` field used in the cross-validation (Section 6).

Three metrics are reported and are not interchangeable; each claim in this paper names which one supports it.

Floored. $|\text{ref} - \text{mesh}| / \max(|\text{ref}|, |\text{mesh}|)$, over points where $|\text{ref}| \geq 10^{-3}$ of the local (six-component) field peak. This is the relative per-component error of record for the electric field. For $\gamma = 10$ every B point is excluded by the floor: $|\mathbf{B}|$ is about 10^{-10} of the E-dominated local field, far below the 10^{-3} threshold, so the floored metric reports no magnetic accuracy at all; the floored column of Table 3 is therefore electric-only at $\gamma = 10$.

Unfloored. The same ratio at every point. It is the only one that reaches the magnetic field, because the floor excludes it.

Absolute. $|\text{ref} - \text{mesh}| / \max_{\text{point}} |\text{ref}_c|$, the peak-normalized error; it supports whole-field statements.

The judge metric $|\text{ref} - \text{mesh}|/(1 + \max(|\text{ref}|, \text{mesh}|))$ is absolute-dominated: for fields of order 10^{-9} V/m the $+1$ denominator swamps the error, so it reports a crossover at $d = 2$ for every case and does not discriminate.

5.2 Error budget

Table 1 separates the measured error into its five sources, with the measured magnitude of each and the section that carries its full statement.

The mesh/deposit term dominates in the far field at moderate γ and is the only term that the mesh alone can reduce. The velocity-binning term is at or below 10^{-5} for the measured spreads at $B = 16$ and does not enter the single-velocity scope. The near-source quadrature term is the measured limit of the fast near backend in the oblique moving case (Section 5.6); the stationary and axial modes reach 10^{-3}. The far-tail term is bounded by the tolerance-level margin of Section 3.7 and is 1.9×10^{-3} of that level at the switch radius. Output precision is not a measured loss: the float32 checkpoint arrives bitwise in PIConGPU, and the float64 path is a uniform opt-in.

Shape bias. The deposited shape is a deliberate smoothing of the point field, not a mesh error. For a stationary point charge the relative bias at 2, 4 and 8 cells is -8.31×10^{-3}, -5.62×10^{-4}, -3.55×10^{-5} for CIC; -1.20×10^{-2}, -8.41×10^{-4}, -5.33×10^{-5} for TSC; and -1.55×10^{-2}, -1.12×10^{-3}, -7.10×10^{-5} for PQS. TSC is the closest of the three to the point field at these distances. The shape option exists to match the PIConGPU species, not to reduce this bias.

5.3 Reference-separated error at 48 cells

Table 2 shows two things. At $\gamma = 10$, aspect 1, the mesh against the point field ($M - P = 1.5 \times 10^{-2}$) is about 58× the mesh against the TSC-convolved continuum ($M - T = 2.6 \times 10^{-4}$, an L2 ratio); the difference is the TSC smoothing, which $M - P$ inherits and $M - T$ does not. A mesh-versus-point comparison measures the shape the deposit applies, not the discretization error. Second, the $M - T$ error is not converged perpendicular to the beam at $\gamma h \gg 1$: at aspect 1 it is acceptable only for $\gamma = 10$ and not for $\gamma = 100$; at aspect 4 it is not converged at any γ. Only the axial direction is clean at every γ.

5.4 Far field

Figure 1 plots the mesh against the TSC-convolved continuum. The measured value is 1.1×10^{-4} at $d = 12$ for $\gamma = 1$, aspect 1, and 2.6×10^{-4} at $d = 32$ for $\gamma = 1$, aspect 4; at $\gamma = 10$, aspect 1 the relative

Table 1: Error budget. Each term is the measured contribution of one error source, with the measurement that establishes it. "Not measured" marks a term whose input data are not in the measurement campaign.

source	measured magnitude	statement
mesh / deposit	1.38×10^{-7} (M–T, $\gamma=1$, aspect 1, $d=48$); 2.6×10^{-4} ($\gamma=10$, aspect 1, $d=48$)	Table 2; far-field sweeps of Section 5.4
velocity binning	4.3×10^{-6} relative ($B=16$ vs 64, 5% energy spread); 0.004% in B_z for the 5% energy case; 1.4% worst against the arbiter for the 1 mrad case	Appendix D, Table 18
near-source quadrature	7.99×10^{-4} stationary interior; 1.69×10^{-4} moving axial interior; 8.15×10^{-2} moving oblique interior (measured limit)	Table 6
far tail	8.53×10^{-2} V/m at the $6R$ switch, 1.87×10^{-3} of the tolerance level; 2.8×10^{-3} V/m worst on the 512^3 output	Appendix D.1, Table 19
output precision	float32 checkpoint: bitwise arrival under PIConGPU, $\max_{\text{abs diff}}=0.0$ after one step; float64 opt-in records a warning attribute	Section 6

error is 6.5×10^{-3} at $d=32$ and 1.8×10^{-3} at $d=48$. These are floored electric values; the unfloored magnetic error at $\gamma=10$ is 9.7×10^{-4} at $d=32$. The window starts at $d=2$, so the TSC-support interior ($|s|<1.5$ cells) is not covered by these sweeps; it is what the near shell measures (Section 5.6). The axial component is below 3×10^{-4} from $d=6$ for every γ.

Table 4 gives the first distance at which every larger distance is at or below the gate, per axis direction.

At $\gamma=1$, aspect 1, the mesh is within 3×10^{-4} of both references from $d=8$; the two are interchangeable there. From $\gamma=10$ up, against the point field the measured perpendicular error at $d=48$ is 6.5×10^{-3} ($\gamma=10$, aspect 1); against the TSC-convolved continuum the perpendicular crossover is $d=48$ ($\gamma=10$, aspect 1), and for the remaining configurations the measured value at $d=48$ exceeds the gate. The axial direction is 6 against the continuum at every γ.

5.5 The γ landscape: 10 to 100

The error landscape for a single particle at $\gamma=10,50,100$ is measured on a 96-cell box, cell 10^{-7} m, aspect 1, axial, at $h=1$ and $h=1/4$ against the independent continuum. The metric is the peak-normalised absolute error per component at $d=8$ cells for the h-comparison.

The error grows steeply with γ: 4.4×10^{-2} at $\gamma=10$, 1.05 at $\gamma=50$, 5.47 at $\gamma=100$, at $h=1$ and $d=8$ cells. Refining to $h=1/4$ brings these to 6.9×10^{-5}, 5.6×10^{-2}, and 0.82. The controlling length is the beaming cone width d/γ: for $\gamma\geq50$ the cone is sub-cell, the h-refinement exponent p collapses to about 1, and refinement stops paying off. The h targets for 10^{-2} derive from the dominant $d=2$ near-charge band: the oblique (45°) target is $h\leq0.002$ at $\gamma=50$ and the axial (0°) target is $h\leq0.001$ at $\gamma=50$ (5.6×10^{8} times the nodes). For $\gamma\geq50$ the cone is sub-cell and 10^{-2} needs $h\leq0.002$ ($\geq10^{8}$ times the nodes); 10^{-3} is out of reach ($>10^{9}$ to 10^{12} times the nodes). These are measured limits of the mesh at a fixed distance in cells.

Table 2: L2-pooled relative error at $d = 48$ cells, $h = 1$, perpendicular direction (L2 over the six components; single-component worst: $\gamma = 1$, aspect 4, $M-T$ 2.65×10^{-5}; $\gamma = 10$, aspect 1, 1.4×10^{-3}). $M-T$: mesh vs TSC-convolved continuum; $M-P$: mesh vs point field; $T-P$: TSC smoothing, mesh-independent. At $\gamma = 10$, aspect 1 the $M-P$ vs $M-T$ gap is the smoothing (L2 ratio 58×), not the mesh.

γ	aspect	$M-T$	$M-P$	$T-P$
1	1	1.38×10^{-7}	9.89×10^{-8}	3.96×10^{-8}
1	4	2.65×10^{-5}	2.39×10^{-3}	2.40×10^{-3}
10	1	2.6×10^{-4}	1.5×10^{-2}	1.48×10^{-2}
10	4	3.33×10^{-2}	1.19×10^{-1}	1.44×10^{-1}
100	1	3.75×10^{-1}	2.20×10^{-1}	4.40×10^{-1}
100	4	2.78×10^{0}	2.49×10^{-1}	8.32×10^{-1}

Table 3: Worst per-component relative error (over x, y, $|\text{ref}| \geq 10^{-3}$ of the local field) of the CLI against the TSC-convolved continuum, at $d = 16$ and $d = 32$ cells; `rosi_v100`, the order-12 independent continuum reference (refine 3). "s/c" = single/cloud.

γ, aspect	s/c	d=16	d=32
1, 1	3.5e−5 / 1.0e−4	2.2e−6 / 2.0e−5	
1, 4	2.8e−3 / 2.1e−3	2.0e−4 / 2.6e−4	
10, 1	5.6e−2 / 5.3e−2	6.3e−3 / 6.5e−3	
10, 4	5.5e−2 / 5.0e−2	6.3e−3 / 6.4e−3	

5.6 The fast near evaluator, per component, by region

Table 6 gives the domain of validity of each near-shell mode. The metric is $\max|\text{mesh}-\text{ref}|/(1+\max|\text{ref}|)$ over 20 interior, 16 face and 16 outside points, cells 10^{-7} m. The per-component values behind the stationary interior row are 7.99×10^{-4}, 3.55×10^{-4}, 6.05×10^{-4}, 0, 0, 0 for $E_x, E_y, E_z, B_x, B_y, B_z$ (the magnetic field is identically zero for a stationary bunch); the moving axial interior row is 1.69×10^{-4}, 1.68×10^{-4}, 1.36×10^{-5}, 1.67×10^{-4}, 1.69×10^{-4}, 0; the moving oblique interior row is 2.43×10^{-2}, 7.42×10^{-2}, 6.32×10^{-2}, 8.15×10^{-2}, 2.96×10^{-2}, 3.89×10^{-2}.

The validity of each mode follows from the table. `exact` is the adaptive FP64 quadrature shared with the reference driver. Its in-support moving branch integrates in Lorentz-contracted coordinates, where the Liénard kernel becomes a Coulomb kernel and the beaming anisotropy moves into a smooth TSC density; the pyramid singularity cancellation is then exact for any velocity. Against the order-12/refine-3 arbiter at $\gamma = 10$ it reaches about 10^{-4} at the axial points $(1, 0, 0)$ and $(0, 0.5, 0)$, about 10^{-3} for the oblique points, and about 3% at the closest axial node. It is therefore a usable reference, but the closest node is not a blanket 10^{-3} path. `fast` meets 10^{-3} in every region for the stationary and moving axial modes; its measured limit is the moving oblique interior, 8.15×10^{-2} at the default order. Against the order-12/refine-3 continuum arbiter the fast backend is 1.6% at the closest moving in-support node. The older driver panels in Appendix C predate the Lorentz-contracted reference and are kept as an upper bound.

The shell radius and order are the cost levers (Section 5.7). The fast evaluator runs the same algebra on the CUDA device when the build has one and a device is present (`-near-shell fast`); `fast-gpu` is the explicit device request. The kernel uses one thread per (pair, component) and agrees with the host path to 5.4×10^{-16} to 8.0×10^{-16}. On `rosi_v100` (64^3 output, $R = 3$, tile 4, 2000 particles, 8.77×10^6 evaluations) the device path runs in 2.5 s against 18.3 s on the host, a 7.3× speedup; it falls back to the host with a warning when no device is present.

5.7 Cost at 2×10^6 particles, one H100

The near shell is the dominant term in every shell row; the mesh-only wall is 2.48 s, of which the compute is 0.116 s (deposit 34.8–40.5 ms, kernel 1.1 ms, FFT 7.3–7.6 ms, read-out 8.0–8.5 ms, writer 48.8–254.5 ms). The support-only shell costs 281.0 s end-to-end at 7.99×10^{-4} interior error, the three-cell shell 955.4 s, and the four-cell shell 1510.7 s, at the same interior error. $R = 1.5$ to $R = 4$ multiplies the end-to-end wall by 5.4× at fixed particle count and grid, while the accuracy of the in-shell region is already at 7.99×10^{-4} at $R = 1.5$. The shell rate on the integrated path is 1.55×10^6, 1.61×10^6 and 1.76×10^6

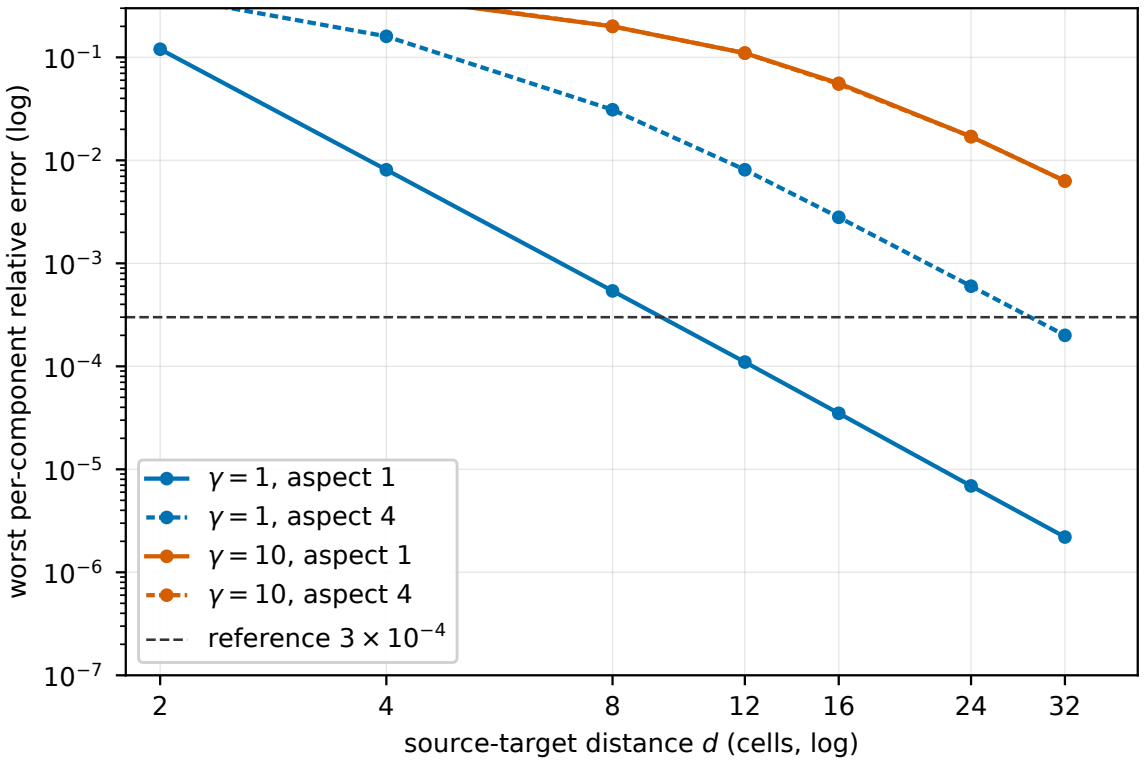


Figure 1: Worst per-component relative error against source-target distance d for the single-particle floored electric series on `rosi_v100`, both axes logarithmic; hue is γ (blue $\gamma = 1$, vermillion $\gamma = 10$) and line style is aspect (solid 1, dashed 4), against the order-12 independent continuum reference (refine 3). The error falls from 1.2×10^{-1} at $d = 2$ to 2.2×10^{-6} at $d = 32$ for $\gamma = 1$, aspect 1. The three-particle cloud series is in Table 3; the $\gamma = 10$ point at $d = 48$ (1.8×10^{-3}) is in the same table but outside the plotted window.

pairs/s for the three radii, about 2× slower per component than the standalone kernel, the overhead of the per-tile local FFT and launch in the integrated path. A distributed FFT decomposes the output field in z slabs, one slab per rank. The transform and transpose agree with the replicated transform to 2.2×10^{-16}. The end-to-end field, including the device TSC deposit into the interleaved complex slab, matches the replicated field to 3.2×10^{-16} at one rank and 5.4×10^{-16} at four ranks in the written file on HIP+MPI (hal); the same four-rank case on CUDA+MPI (jureca-h100, four H100s) gives 2.5×10^{-16}. The host FFTW backend supports strided plans in the current alpakaVendor and matches the replicated field to 1.9×10^{-16} at two MPI ranks. The automatic policy selects the slab path above its padded-node threshold. At a 640^3 output the per-rank host memory is 41.1 GB at 2 ranks and 20.6 GB at 4, the quantity the decomposition buys: the output field is slab-local, while the bunch remains replicated on every rank. The `MPI_Alltoallv` transposes are 74–85% of the FFT-plus-transpose time, so the distributed path is a memory feature rather than a speed feature.

5.8 Determinism

Repeated runs of the CLI are bit-identical: two runs of the same input produce byte-identical E/B mesh payloads (md5 `d3641d64...`, 15 092 736 bytes). The CLI also rejects a below-floor `-tolerance` (10^{-15}) with a nonzero exit. The `gridUnitSI` is applied once, in the reader, and the mesh geometry follows `gridSpacing` · `gridUnitSI`; a mesh without a positive `gridUnitSI` is rejected.

5.9 Anisotropic cells

At anisotropic spacing the accuracy is quantified per geometry, not inherited from the cubic case. On a grid with per-axis cell spacing $(1, 1, 4)$ the stretched z axis matches the continuum to $\approx 10^{-3}$–10^{-7} across the in-grid distances, the same scale as the cubic control; the short x/y axes sit in the near-field TSC regime at a short physical distance ($d = 4$ cells is only 4 m along x/y, so $E_z \approx 0.19$ stationary and

Table 4: Crossover radius in cells: first d at which every larger d is at or below the gate, $h = 1$, L2 over the six components, per axis direction, worst over the three positive axis directions (>48: beyond the measured window). T: TSC-convolved continuum; P: point field. Oblique directions are $\sim 1.5\times$ larger.

	3×10^{-4}		10^{-3}			
γ	aspect	ref	axial z	perp x	axial z	perp x
1	1	T	8	8	6	6
	1	P	8	8	6	6
1	4	T	6	24	6	16
	4	P	48	> 48	32	> 48
10	1	T	6	48	6	48
	1	P	> 48	> 48	32	> 48
10	4	T	6	> 48	6	> 48
	4	P	> 48	> 48	32	> 48
100	1	T	6	> 48	6	> 48
	1	P	> 48	> 48	32	> 48
100	4	T	6	> 48	6	> 48
	4	P	> 48	> 48	32	> 48

Table 5: Single-particle mesh error at $d = 8$ cells, 96-cell box, cell 10^{-7} m, aspect 1, axial, `jureca-h100`. Peak-normalised absolute error, worst per component, at $h = 1$ and $h = 1/4$. The h-refinement exponent p (error $\sim h^p$, fitted from the two spacings) falls from 1.85 to about 1.0 as γ rises.

γ	$h = 1$	$h = 1/4$	slope p	note
10	4.4×10^{-2}	6.9×10^{-5}	1.85	
50	1.05	5.6×10^{-2}	1.06	$h = 1$ off by more than the peak
100	5.47	0.82	1.01	$h = 1$ off by more than the peak

E_x/E_y up to ≈ 0.9 at $d = 4$ for the $\gamma = 10$ case). The mesh is not wrong because the spacings differ; the near points on the short axes are physically closer to the source.

6 Cross-validation and checkpoint transport

Two independent checks constrain the method externally: a cross-validation against PIConGPU's exact-quadrature field reference, and a checkpoint and restart transport. Neither is a truth comparison.

6.1 Cross-validation against the PIConGPU reference

The reference is the TSC-convolved field (Section 5), evaluated in PIConGPU [Burau et al., 2010] in its reference-only field-initializer mode. The cross-validation suite passes 10/10, and both sides use identical Yee offsets, grid, and origin; the reference staggered point maps to a picBunch node by index with zero snap deviation.

At 318 staggered points on a $64 \times 64 \times 32$ grid, cell 10^{-7} m, the overall worst per-component relative error is 4.86×10^{-2} for $\gamma = 1$ (E/z) and 6.20×10^{-1} for $\gamma = 10$ (E/z; E/y is 5.16×10^{-1}), against a reference reproducibility floor of 1.19×10^{-6} (E) and 1.43×10^{-6} (B). picBunch is about four to six orders of magnitude above that floor; the measured error is concentrated at the core and the beamed peaks, where the TSC support is, while the bulk agrees to below 1–3% and B is exactly zero for $\gamma = 1$ in both components.

The whole-field comparison, $N = 1000$ on $32 \times 32 \times 16$, cell 2×10^{-7} m, 16384 nodes per component, gives the same picture: $\gamma = 1$ bulk median 3.8–7.8×10^{-3} with an L2 of 4.6–5.7×10^{-2}; $\gamma = 10$ bulk median 1.4–3.3×10^{-2} but a beamed-core overshoot that pushes the E/y, E/z L2 to 0.76–0.92. The forward E/x is well resolved at $\gamma = 10$ ($\sim 6\%$); the beamed transverse E/y, E/z are 50–130% worst at the core, where the Lorentz-contracted field is a sub-cell pancake below the TSC support and the spectral near-field kernel resolution.

The disagreement is per-source, local to the TSC support, and independent of particle count: on a $64 \times 64 \times 32$, cell 10^{-7} m, stationary bunch ($\gamma = 1$) the band is flat in N from 1 to 1000 (10^{-3}–3×10^{-2}). The $O(1)$ values at $N \geq 10$ occur only because the global worst sample point falls 0.415–0.65 cells from

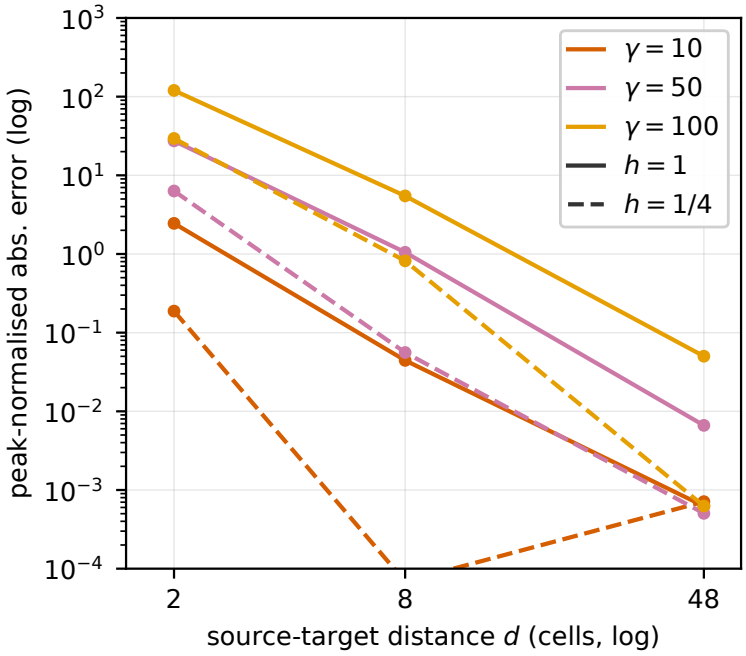


Figure 2: Peak-normalised absolute error against distance on log-log axes, single particle, 96-cell box, cell 10^{-7} m, aspect 1, axial, `jureca-h100`. Solid/dashed is $h = 1/1/4$ and hue is $\gamma = 10$ (vermillion), 50 (purple), 100 (orange). At $h = 1$ the error exceeds the peak field by $d = 8$ cells for $\gamma = 50$ and $\gamma = 100$.

Table 6: Fast near evaluator, worst per-component relative error against the order-12 independent continuum reference, cells 10^{-7} m, `jrc0880` (`sm_90`). Regions: interior (inside the TSC support), face (on its boundary), outside (1.6–3.6 cells). The "remainder order" column is the Gauss–Legendre order of the smooth remainder; the singular asinh box is O(1) in every row (moving axial uses the γ-stretched box, moving oblique the rest-frame octant box at order 8).

γ, direction	remainder order	interior	face	outside
1, stationary	5	7.99×10^{-4}	1.96×10^{-5}	1.57×10^{-7}
10, axial	8	1.69×10^{-4}	1.11×10^{-4}	2.12×10^{-6}
10, oblique	5	8.15×10^{-2}	3.94×10^{-3}	1.01×10^{-4}

a source, inside that source's TSC support; an incorrect charge weighting would grow with N, and it does not.

6.2 Checkpoint format and bitwise arrival

The checkpoint writer emits a PIConGPU openPMD field checkpoint matching PIConGPU's own plugin schema: root layout, openPMD version 5.0 [Huebl et al., 2015], E, B $x/y/z$ at the exact Yee positions, grid spacing and offset, and the `picongpu_idProvider` mesh metadata. The default precision is uniform `float32` for every field and particle record, so a stock single-precision PIConGPU can load the file; `float64` is a uniform opt-in that records a warning attribute because stock PIConGPU cannot load double records. Every species from the input bunch is copied, the `charge` and `mass` records are written rather than omitted, and the species is a single particle patch, `patch` = `"bbox"` or `"domain"`. For a multi-rank restart the writer splits the species into one patch per output rank: `patch_count` in `[placement]` gives cumulative, disjoint `numParticlesOffset` ranges, with one-cell overlap between neighbouring z regions, and every particle belongs to exactly one patch.

Field arrival is bitwise. A fresh checkpoint written from a two-species input (16 electrons and 16 ions, `picongpu_idProvider` startId 0, nextId 32) restarts in PIConGPU with exit 0 under both `patch` modes (`bbox`: offset $(7, 8, 3)$, extent $(8, 10, 6)$; `domain`: offset $(0, 0, 0)$, extent $(24, 24, 12)$), all six field components bitwise-equal, $\max_{\text{abs diff}} = 0.0$, electrons with ids 0 to 15, first-step displacement exactly $\beta c\, dt$ with $\beta = 0.9949874367$. A multi-rank restart with the four-patch layout was verified process-level: a 256-particle bunch written as four non-empty patches (cumulative offsets $0, 56, 126, 203$, one-cell

Table 7: Measured CLI wall at 2×10^6 particles, 20-cell bunch, $66 \times 68 \times 66$, cell 2.5×10^{-7} m, tile 4, one H100 (`jrc0880`, `sm_90`). "near-shell wall" excludes the mesh, "end-to-end wall" includes it. The mesh-only compute inside its 2.48 s wall is 0.116 s.

configuration	near-shell wall (s)	end-to-end wall (s)	worst near error
mesh only	0	2.48	no near correction
support-only shell, $R = 1.5$, order 5	278.28	281.0	7.99×10^{-4}
three-cell shell, $R = 3$, order 5	908.14	955.4	7.99×10^{-4} in, $\leq 1.6 \times 10^{-7}$ out
four-cell shell, $R = 4$, order 5	1508.12	1510.7	7.99×10^{-4} in, $\leq 1.6 \times 10^{-7}$ out

Table 8: Whole-field comparison of picBunch against the PIConGPU `referenceOnly` field, $N = 1000$, $32 \times 32 \times 16$, cell 2×10^{-7} m, 16384 nodes/component, rosi_v100. `worst_rel` is peak-normalized; median and p99 are per-point over nodes with $|\text{ref}| > 10^{-3} \times$peak; L2 is pooled. Not a truth comparison.

component	worst_rel	median	p99	L2
$\gamma = 1$ (B identically 0)				
E/x	1.48×10^{-1}	3.92×10^{-3}	3.20×10^{-1}	4.62×10^{-2}
E/y	1.49×10^{-1}	3.76×10^{-3}	2.93×10^{-1}	4.75×10^{-2}
E/z	1.22×10^{-1}	7.81×10^{-3}	3.65×10^{-1}	5.67×10^{-2}
$\gamma = 10$ (B/x identically 0)				
E/x	1.36×10^{-1}	4.80×10^{-3}	1.23×10^{-1}	4.44×10^{-2}
E/y	1.98	1.36×10^{-2}	7.25	7.63×10^{-1}
E/z	1.93	3.33×10^{-2}	10.1	9.19×10^{-1}
B/y	4.42×10^{-1}	3.12×10^{-2}	1.41	2.58×10^{-1}
B/z	4.63×10^{-1}	1.52×10^{-2}	0.93	2.20×10^{-1}

overlap) restarts PIConGPU with exit 0, and after one step all six field components are bitwise-equal to the checkpoint, $\max_{\text{abs diff}} = 0.0$.

Placement and particle-data integrity. A 12-marker bunch with an asymmetric position and momentum pattern was placed at the xz middle and the y first quarter of a $32 \times 64 \times 32$ domain. picBunch reports centroid fractions $(0.500000, 0.250000, 0.500000)$; after one PIConGPU step the centroid is $(0.500755, 0.250446, 0.501466)$, the raw momentum records are preserved exactly, and the per-particle displacement matches $dt\,c\,u/\gamma$ to 4.1×10^{-7} cells. The displacement-to-momentum ratio is one positive scalar on all three axes (spread 8.9×10^{-6}), so the restart has no axis permutation and no mirror. One metadata requirement is sharp: PIConGPU restart ignores the particle `unitSI` and interprets momentum in `sim.unit.mass()`·`unit_speed`, so a checkpoint written with the electron mass as `unit_mass` restarts with all momenta 10^4 too large; `unit_mass` must equal the deck's `sim.unit.mass()`.

On a $24 \times 24 \times 12$ deck the restart is exit 0 with `-periodic 1 1 1`, and the 16 electrons load; initialization drops from the ~ 116 s neuralBunch path to 1.8 s because the checkpoint carries the field. For an open-boundary deck the writer copies the `Convolutional PML` slabs from the input checkpoint verbatim (converting a non-float32 slab to float32), so the PML state arrives with the restart. On a real LWFA checkpoint (638976 electrons, eight `Convolutional PML` groups with six datasets each, $64 \times 128 \times 64$, `-periodic 0 0 1`) the writer copies all eight groups and the restart exits 0; the copied PML datasets are float32 and bitwise equal to the source, initialization is 3.986 s, and the main E and B records are recomputed from the particles and stay float32.

7 Performance and scaling

All compute timings are on one host, `rosi_v100` (node `gv002`), exclusive Slurm allocation, pinned to GPU 0. Build: CUDA 12.6, `PICBUNCH_ARCH=70`, Release, padding 2, double. Each configuration ran three timed repeats after one untimed warm-up plus one device-level warm-up; stage values are medians, the total is min/median/max. The four $\gamma = 10$, $N = 2 \times 10^6$ repeats give total medians 60.35 to 64.76 ms, about 7% scatter; the 64.81 ms row is the $\gamma = 100$ run. These compute-only totals exclude the checkpoint writer; the end-to-end totals with the writer are in Table 10, a different measurement path.

Table 9: Per-stage medians (ms) and device allocation (MiB), padding 2, double, `rosi_v100` GPU 0. "box" is the source-box size in cells; M is the padded FFT array per axis. Stage columns are medians over the three timed repeats.

N	box	γ	h	n	M	total	deposit	kernel	FFT	read-out	dev
10^3	20	10	1	30	60	10.17	1.13	0.48	4.36	4.08	8.4
10^4	20	10	1	30	60	10.69	2.42	0.53	4.37	3.35	8.7
10^5	20	10	1	30	60	12.90	4.59	0.48	4.07	3.28	11.4
10^6	20	10	1	30	60	61.92	53.61	0.53	4.31	3.45	38.9
$2{\cdot}10^6$	20	10	1	30	60	64.76	56.51	0.43	3.80	3.69	69.4
$2{\cdot}10^6$	50	10	1	60	120	85.67	52.82	2.85	13.89	15.20	127.4
$2{\cdot}10^6$	100	10	1	110	220	232.97	51.89	22.66	75.97	82.40	468.7
$2{\cdot}10^6$	20	10	1/2	50	100	69.27	52.38	1.29	7.43	8.16	99.5
$2{\cdot}10^6$	20	10	1/4	90	180	151.51	50.94	11.76	43.38	44.58	284.5

Table 10: End-to-end pipeline time including the checkpoint writer and the openPMD reader, medians over 3 repeats, `rosi_v100` GPU 0, 20-cell box, $\gamma = 10$, CLI `-tolerance` 10^{-3}. A different measurement path from the compute-only Table 9: the totals include the writer (the dominant added term); the per-stage split is deposit/kernel/FFT/readout/writer.

N	total	deposit	kernel	FFT	readout	write
10^4	22.00	2.70	0.53	4.53	3.06	11.29
10^5	24.87	5.82	0.56	4.31	3.80	10.65
10^6	47.15	31.56	0.46	4.20	3.95	7.54

Reproducibility. The matrix of Table 9 is the `picBunchBench` sweep, run with picBunchBench--sweep --repeat3--csvmatrix.csv. The input bunches are not checked into the repository; the sweep configuration is the generator. The fast GPU kernel tools/closed-form/gpu_extract.cu is part of the source. The H100 runs used CUDA 13.4.

Correctness anchor. Every timing run is anchored by the Coulomb limit ($\mathbf{u} = 0$): one particle at a node, far nodes at $d = 8, 16, 24, 32$ along all six axis directions. The mesh field against the direct TSC stencil sum of the analytic Coulomb kernel has worst relative error 4.88×10^{-14} (max absolute 2.28×10^{-25}), passing the 10^{-12} threshold. This confirms that the zero-padded FFT convolution reproduces the TSC-convolved Coulomb field to roundoff (Section 4).

Figure 3 sweeps one axis at a time: the top panel fixes the 20-cell box and varies N, the bottom panel fixes $N = 2{\cdot}10^6$ and varies the box. The scaling is in two regimes, with the crossover set by the box, not the particle count. At large N the deposit dominates, linear in N, 52–57 ms on the 2×10^6 rows while the FFT, kernel, and read-out stay under 10 ms at the 20-cell box; the total is flat across $\gamma = 1, 10, 100$ (64.9/64.4/64.8 ms). At large boxes the kernel and FFT scale with M^3: from $M = 60$ to $M = 220$ the total goes from 64.8 ms to 233 ms, and halving h costs 1.07× and 2.35× because the deposit stays constant. The per-h totals are 64.76, 69.27, and 151.51 ms for $h = 1$, 1/2, 1/4 (padded array $M = 60/100/180$, 2×10^6 particles, 20-cell box). On the H100 (`jrc0880`, `sm_90`), the same 2×10^6, 20-cell, $h = 1$ mesh is 0.052 s compute (deposit 34.4 ms, kernel 1.08 ms, FFT 7.44 ms, read-out 9.02 ms; medians of three), and 0.755 s CLI wall including the particle read and the HDF5 write.

No configuration came near the 32 GB device limit; the largest exact device allocation was 468.7 MiB. Memory scales with the padded bunch box, not the global field.

Table 10 is the same pipeline with the writer in the loop. The deposit is still linear in N (2.70 → 31.56 ms from 10^4 to 10^6), and the FFT is still box-bound and flat. The writer is the dominant added term: 11.29 ms at 10^4 and 7.54 ms at 10^6. The $N = 10^6$ total is 47.15 ms with the writer, versus 61.92 ms in the compute-only Table 9; the two campaigns differ in measurement path and timing boundary, so neither is the other's total plus the writer.

7.1 MPI rank scaling

The rank-parallel compute (Section 3.8) and the chunked writer were measured on an idle `rosi_v100` window, medians of three runs. The node was checked idle before the block (no CPU worker, 1-min load

Table 11: Rank scaling on `rosi_v100`, idle window, medians of three runs. Mesh: $N = 2 \times 10^6$, 112^3, component-parallel (replicated deposit and source FFT). Near shell: $N = 128$, 32^3, $R = 1.5$, tile 8, tile-parallel. Write: 256^3, 0.80 GB, collective HDF5. Mesh and near shell are bitwise identical at 1/2/4 ranks.

	near shell			mesh		write	
ranks	wall (s)	speedup	eff.	wall (ms)	speedup	wall (s)	speedup
1	36.61	1.00	100%	522.2	1.00	1.26	1.00
2	29.60	1.24	62%	436.3	1.20	1.33	0.95
4	27.29	1.34	34%	433.6	1.20	1.43	0.88

Table 12: H100 remeasurement, medians of three serialized runs. Top: $N = 2 \times 10^6$, one bin, one rank, stage times in ms. Bottom: 256^3 output, $N = 2 \times 10^6$, total, writer and read-out in ms.

box	deposit	FFT	read-out	writer	total
20 cells	36.1	1.53	17.5	5.1	60.5
100 cells	38.9	31.3	21.4	53.8	147.9

256^3	1 rank	2 ranks	4 ranks
total (ms)	3899	3737	3664
writer (ms)	3319	3327	3323
read-out (ms)	204	106	44

below 2), but each rank block starts from the preceding measurement, so the 1-min load rises within the block: the near-shell block starts at 1.9, 8.4 and 11.8 for 1, 2 and 4 ranks (mesh block 1.4–1.5, write block 1.1–1.6); no foreign process was active during the window. Correctness was verified at 1, 2 and 4 ranks on a 128-particle, 32^3, cell 8×10^{-7} m, tolerance 10^{-3}, near-shell $R = 1.5$, tile 8 problem: the mesh and the near shell are bitwise identical at 1-vs-2 and 1-vs-4, all six components, $\max_{\text{abs diff}} = 0.0$.

Figure 4 shows the speedup for the three parallel pieces. The mesh is component-parallel: the replicated deposit (307–332 ms, flat in rank count) and the replicated source-spectrum FFT cap it, so the speedup plateaus at 1.20× at both 2 and 4 ranks. The near shell is tile-parallel: the static round-robin tile split plus the all-reduce of the full correction array leave static-tile imbalance that plateaus it at 1.34× at 4 ranks. The writer is a collective HDF5 write of one shared series: on a 256^3 (0.80 GB) output it takes 1.26, 1.33 and 1.43 s at 1, 2 and 4 ranks (0.88× at four ranks), so the parallel writer stays within 13% and does not become the bottleneck at this size.

7.2 H100 remeasurement

A fresh campaign on `jureca-h100` (four H100 PCIe 80 GB, sm_90, CUDA 13.4, serialized on one GPU, medians of three runs) reproduces the scaling picture with current hardware. At $N = 2 \times 10^6$, a 20-cell box takes 60.5 ms end to end (deposit 36.1, FFT 1.53, read-out 17.5, writer 5.1 ms); a 100-cell box takes 147.9 ms (deposit 38.9, FFT 31.3, read-out 21.4, writer 53.8 ms). The deposit remains linear in N; the 100-cell case is FFT- and writer-bound. For $N = 200$ particles the 16- and 64-bin multi-velocity runs take 92.2 and 100.6 ms; the B_z peak differs by 4.6%, within the shot noise of 200 particles.

Rank scaling at $N = 2 \times 10^6$ is flat in total time because the shared HDF5 writer dominates: at 256^3 the total is 3899, 3737 and 3664 ms and the writer 3319, 3327 and 3323 ms at 1, 2 and 4 ranks, while the read-out falls from 204 to 44 ms. The writer is the remaining multi-rank bottleneck at this size; the compute stages scale only modestly because the deposit and source spectrum are replicated.

8 Discussion

The method computes the present-time field of a uniformly moving, TSC-shaped bunch as a free-space FFT convolution. It is a far-field method, and the measured error structure says exactly where that stands.

In the far field the relative error against the TSC-convolved continuum is below 3×10^{-4} on the beam axis from $d = 6$ for every γ; perpendicular to the beam the crossover is $d = 8$ for $\gamma = 1$, aspect 1, $d = 32$

for $\gamma = 1$, aspect 4, and $d = 48$ for $\gamma = 10$, aspect 1. The mesh against the point field carries the TSC smoothing in addition to the deposit error, and the two disagree by a factor of 58 in L2 at $\gamma = 10$, aspect 1, $d = 48$ (Table 2); a mesh-versus-point number measures the deposit shape, not the discretization. The cost is two regimes with a crossover set by the box, not the particle count: the deposit is linear in N and dominates at large N (52–57 ms on the 2×10^6 rows), while the FFT and kernel scale with M^3 and dominate at large boxes (64.8 to 233 ms from $M = 60$ to 220). Repeated runs are bit-identical (md5 `d3641d64...`), and the Coulomb limit reproduces the TSC-convolved Coulomb field to 4.88×10^{-14}.

The near field is the limit of the mesh. Where the slice planes cut the bunch core, the mesh-only worst per-component plane error is 0.12–0.415 for $\gamma = 1$ and 0.11–4.60 for $\gamma = 10$ (Appendix C). The opt-in near shell assembles exactly to roundoff, and its tile subtraction is not the limiting term: at the measured $R = 3/4$ radii on the $h = 1$ slice grid the residual is the mesh contribution outside the shell, not the subtraction. The fast backend meets 10^{-3} stationary (7.99×10^{-4} interior) and moving axial (1.69×10^{-4} interior); the moving oblique interior is 8.15×10^{-2}, the measured limit of the fast backend at its default order (Table 6). The shell radius is the dominant cost lever: at 2×10^6 particles the support-only shell is 281.0 s end-to-end at 7.99×10^{-4} and the four-cell shell 1510.7 s at the same interior error, against a mesh-only wall of 2.48 s (Section 5.7).

The γ landscape states the mesh's sub-cell limit. At a fixed distance in cells the error grows steeply with γ: 4.4×10^{-2}, 1.05, 5.47 for $\gamma = 10, 50, 100$ at $d = 8$, $h = 1$ (Table 5, Fig. 2). The controlling length is the beaming cone width d/γ; for $\gamma \geq 50$ the cone is sub-cell and the h-refinement exponent p falls from 1.85 to about 1.0, so refinement stops paying off and 10^{-3} is out of reach ($> 10^9$ to 10^{12} times the nodes). At $\gamma = 50$ to 100 the near-charge error requires a shell of radius of order γ cells; that is the regime where the near shell is the working tool rather than the mesh.

The scope of the validated measurements is a single shared velocity, or a set of binned velocity classes with one convolution per class; the near shell needs no binning because each pair carries its own velocity. The measured binned behaviour is bin-independent to 0.004% for the 5% energy spread and 1.4% against the arbiter for the 1 mrad case (Appendix D). The computed quantity is the TSC-convolved uniformly-moving point-charge field; the convolution is free-space (no boundary condition other than zero fill beyond the far-tail switch); and there is no radiation term, because the motion is uniform. The checkpoint arrives bitwise under a stock PIConGPU, periodic and open-boundary, single- and multi-patch, so the field is usable as the start state of the simulation it is meant to initialize.

9 Conclusion

The method computes the present-time field of a uniformly moving, TSC-shaped bunch as a zero-padded free-space FFT convolution with staggered Yee read-out, driven by a single TOML configuration, and writes a uniform float32 openPMD checkpoint that restarts under a stock PIConGPU, periodic and open-boundary, single- and multi-patch. The implementation is validated on V100 `sm_70`, H100 `sm_90`, and gfx1100 HIP hardware.

The method is a far-field tool. In the far field the mesh is below 3×10^{-4} of the TSC-convolved continuum from $d = 8$ cells for $\gamma = 1$, aspect 1, and from $d = 32$ for $\gamma = 1$, aspect 4; at $\gamma = 10$, aspect 1, the measured values are 6.5×10^{-3} at $d = 32$ and 1.8×10^{-3} at $d = 48$. At 2×10^6 particles the mesh-only CLI wall is 2.48 s on one H100 (compute 0.116 s), deterministic (md5 `d3641d64...`), and the cost is linear in the particle count for the deposit and cubic in the padded grid for the FFT. The opt-in near shell assembles exactly to roundoff and its fast backend meets 10^{-3} stationary and moving axial, with the moving oblique interior at 8.15×10^{-2}, the measured limit. The MPI compute is a component-parallel mesh and a tile-parallel near shell, bitwise identical at 1/2/4 ranks, with the replicated deposit and source FFT capping the mesh speedup at 1.20×; the collective writer stays within 13% from one to four ranks. The measured sub-cell limit at $\gamma \geq 50$ is stated in Section 5.5; in that regime the near shell is the working tool.

Methods

9.1 Bunch and grid configurations

Table 13 lists every measured case with the bunch and the grid, and the section that uses it. The entries are the recorded configuration values of this work. The generation line for the mid-bunch bunches is generate_bunch.py--modelgamma--gammaG--seed42--grid646432--cell-size1e-71e-71e-7--centroid32321

6--sigma666--weightsuniform. The total charge is N times $-e$ for the electron bunches with one unit weight, where $e = 1.602176634 \times 10^{-19}$ C; the kinetic energy is $(\gamma - 1)\, m_e c^2$ with $m_e c^2 = 510.999\,\mathrm{keV}$. The mid-bunch bunches are mono-energetic: the read-back records $\gamma_{\min} = \gamma_{\max} = 1$ for the $\gamma = 1$ bunch and $\gamma_{\min} = \gamma_{\max} = 10$ for the $\gamma = 10$ bunch.

9.2 Command and software versions

The binary is invoked as picBunch--inputbunch.h5--destination<checkpoint>--configparams.toml, with every unit and conversion parameter in the TOML file. The `[picongpu]` section carries `unit_length`, `unit_e`, `unit_b`, `dt`, `step`, `precision`, `species`; `[domain]` carries `cells`, `cell_size`, `origin`; `[placement]` carries `offset_unit` and the `left`/`front`/`top` fractions; `[accuracy]` carries `tolerance` and the near-shell settings (`near_shell none`/`exact`/`fast`/ `fast-gpu`, plus `near_shell_radius` and `near_shell_tile`); `patch_count` in `[placement]` splits the species into one particle patch per output rank for multi-rank restarts. A missing conversion key causes a clear error at start-up. The input bunch is centred by its charge-weighted centroid and placed at the requested offset from the domain origin; a placement leaving the bunch outside the domain is rejected at start-up.

The measurement hosts and toolchains are: `rosi_v100` (node `gv002`, V100 `sm_70`, CUDA 12.6, `PICBUNCH_ARCH=70`, Release); `jrc0880` (H100 `sm_90`, CUDA 13.4); `hal` (NVIDIA A30, `sm_80`, CUDA 13.4.92). The openPMD standard is version 5.0 [Huebl et al., 2015]; the reference PIConGPU is the release whose `referenceOnly` field-initializer mode is used in Section 6. The FFT library is cuFFT in the unnormalized form of the vendor documentation [NVIDIA Corporation, 2024].

Particle shape. The charge deposit uses the PIConGPU species shape selected with `-shape` (`cic`, `tsc` or `pqs`; default `tsc`). The support radii are 1, 1.5 and 2 PIC cells. The shape is not recorded in the openPMD particle data, so it must mirror the simulation's `shape.param`. The shape sets the source smoothing; the FFT kernel remains the sampled point kernel. The near shell currently supports TSC only and rejects CIC/PQS, while the mesh path supports all three.

Chunked particle input. `-max-particle-chunk-gb` (a float, for example 1.5) streams the selected species in chunks of at most that many gigabytes instead of loading it once. Each chunk is read on rank 0, broadcast, computed on the same fixed grid, and added to the output field. The mode requires explicit `-grid`, `-origin` and `-cell-size` and a single velocity class; the global-domain writer is not combined with it yet. On a 200-electron test, 1-particle chunks reproduce the unchunked field to 4.8×10^{-16}.

9.3 Data and code availability

The source is available at `https://github.com/psychocoderHPC/picBunch`. The measurement data referenced by name in this paper are the files recorded in `docs/measurements/` of the repository: `matrix.csv` (the bench sweep behind Table 9), `w4b_rel_error.csv` (Fig. 1, Table 3), `coulomb.csv` (the Coulomb anchor), and the per-case `accuracy_results.json` and `n1b_padding_*.csv` files (Appendix B), with SHA-256 checksums in `MANIFEST.sha256`. The field-slice `.npz` arrays behind Figs. 12–16 and Figs. 17–19 are recorded in the measurement campaign (Appendix C); individual measurements name the artifact they used. The figures are rendered at their print width from the recorded data with the shared style module described in the repository documentation; every figure PDF in `figs/` was regenerated from recorded inputs, and the figure audit records the minimum printed font size and colour-bar label of each.

A Derivations

A.1 The singular box

The in-support integral of Eq. (1) is split by writing the kernel as its divergence plus a smooth remainder. The divergence part is the eight-term asinh box

$$T(A,B,u) = \int_0^A \int_0^B \frac{\mathrm{d}X\,\mathrm{d}Y}{\sqrt{u^2+X^2+Y^2}} = \begin{cases} A\,\mathrm{asinh}(B/A) + B\,\mathrm{asinh}(A/B), & |u| < 10^{-300}, \\ A\,\mathrm{asinh}\big(B/\sqrt{u^2+A^2}\big) + B\,\mathrm{asinh}\big(A/\sqrt{u^2+B^2}\big) & \\ \quad -u\,\arctan\big(AB/(u\sqrt{A^2+u^2+B^2})\big), & \text{otherwise.} \end{cases} \tag{12}$$

with $A, B > 0$ (Eq. (10)). Each box face has 8 corners and 24 such calls per component, so the eight-term box is O(1) per pair. The form is even in u: u is the distance from the field point to the face plane, so a field point beyond a box face gives negative u, and the $|u| \to 0$ branch must fire on $|u|$, not u. The branch at $|u| < 10^{-300}$ is the $u \to 0$ limit, which is finite; it is not a regularization, it is the limit of the second branch as $|u| \to 0$.

The primitive is verified against a converged midpoint trapezoid reference: the double integral of the integrand over the unit box with $u = 1$ is -2.6348 for the converged midpoint trapezoid; the naive trapezoid gives -2.6333, 1.5×10^{-3} low, a trapezoid singularity artefact at the corner where the integrand is not smooth. The primitive is implemented in the fast evaluator header `include/picBunch/detail/FastNear.hpp` and the device reference `tools/closed-form/gpu_extract.cu`.

A.2 The smooth remainder

The remainder integrates $(S - W^*)K$ over the TSC knot panels, where S is the step function of the kernel's support and W^* is the TSC shape subtracted so that the integrand is smooth inside the support. Each axis is split at the TSC knots $(-1.5, -0.5, 0.5, 1.5)$ and at the anchor point s^*, the projection of the source onto the axis; on each panel a plain tensor Gauss–Legendre quadrature is applied. The default order is 5 per axis, 8000 nodes per in-support pair. For a moving bunch the box is γ-stretched for the axial direction: the beaming cone compresses the field into a pancake of width d/γ, and the stretched box matches that compression so the singular part stays O(1). The oblique direction uses a rest-frame octant quadrature of the singular box at order 8 (4096 nodes) plus the same remainder, because the octant structure of the rest-frame kernel does not map to the stretched box.

A.3 The h^3 Riemann factor

The sampled convolution evaluates the continuum integral of Eq. (1) by a Riemann sum whose nodes are spaced h cells apart, so each sample covers a physical volume h^3 (in cell units). The unnormalized FFT round trip scales by M^3, so the read-out applies the factor $h_x h_y h_z/(M_x M_y M_z)$ to convert the sum to the integral. This is the only place where the mesh spacing enters the amplitude; the scale-invariance of Section 3.5 follows from it, together with the homogeneity of degree -2 of the kernel and the cell-unit Yee offsets.

B Seven-velocity single-particle validation

Figures 5–11 show the mesh field for one electron on the final 128^3 grid, cell 5×10^{-8} m over the same 6.4 μm domain, against the independent order-12 continuum reference on the mid-z x-y plane. The seven velocities are $\gamma = 1.0$, 1.1, 1.5, 2, 10, 50 and 100. Each figure is one velocity case, with y horizontal and increasing left to right, and its rows are labelled with γ, β and the particle count $N = 1$. Table 16 lists the measured worst relative errors: 3.474×10^{-3}, 3.669×10^{-3}, 3.719×10^{-3}, 5.098×10^{-3}, 1.197×10^{-1}, 1.426 and 3.213 in velocity order. The error grows with γ because the field is compressed into a fraction of a cell: at $\gamma = 50$ and $\gamma = 100$ the Lorentz pancake is thinner than the mesh spacing, so those cases show the geometric sub-cell limit of the mesh, not a solver error. Raising the resolution from 64^3 to 128^3 at the same physical domain left the relative errors unchanged, so the growth with γ is the field compression rather than a discretization that finer cells remove. The $\sim 3.5 \times 10^{-3}$ floor at low γ is the

reference/quadrature floor at the settings used, not the solver's limit. The peak squared fields scale as $1/\text{cell}^2$ compared with the 64^3 series (a factor 16 for the halved cell), a consistency check on the two campaigns.

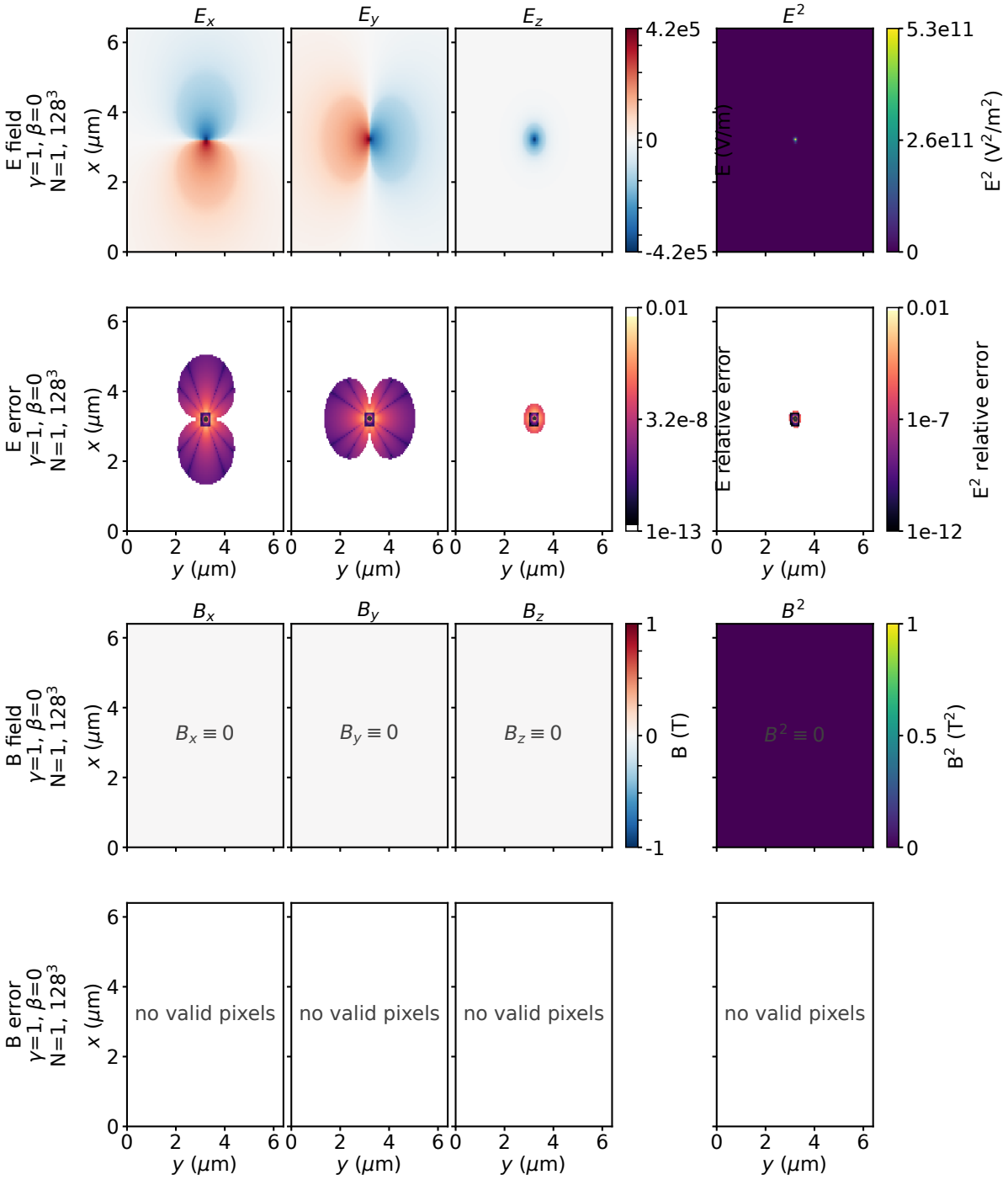


Figure 5: Single-electron field and error on the x-y plane at mid-z for $\beta = 0$ ($\gamma = 1.0$) on the 128^3 grid, cell 5×10^{-8} m. The bunch moves along $+y$: y is horizontal and increases to the right, x is vertical. The E block and the B block each put E_x, E_y, E_z (or B_x, B_y, B_z) on one shared colour scale with its own bar and E^2 (or B^2) on a separate colour scale with its own bar, so the squared-magnitude scale never dominates the component scale. Error rows are logarithmic, carry the 10^{-3} contour, and are masked below 10^{-3} of the reference peak; at $\gamma = 1.0$ the B error bars are omitted because no pixel exceeds the mask threshold; the worst component error is 3.474×10^{-3}.

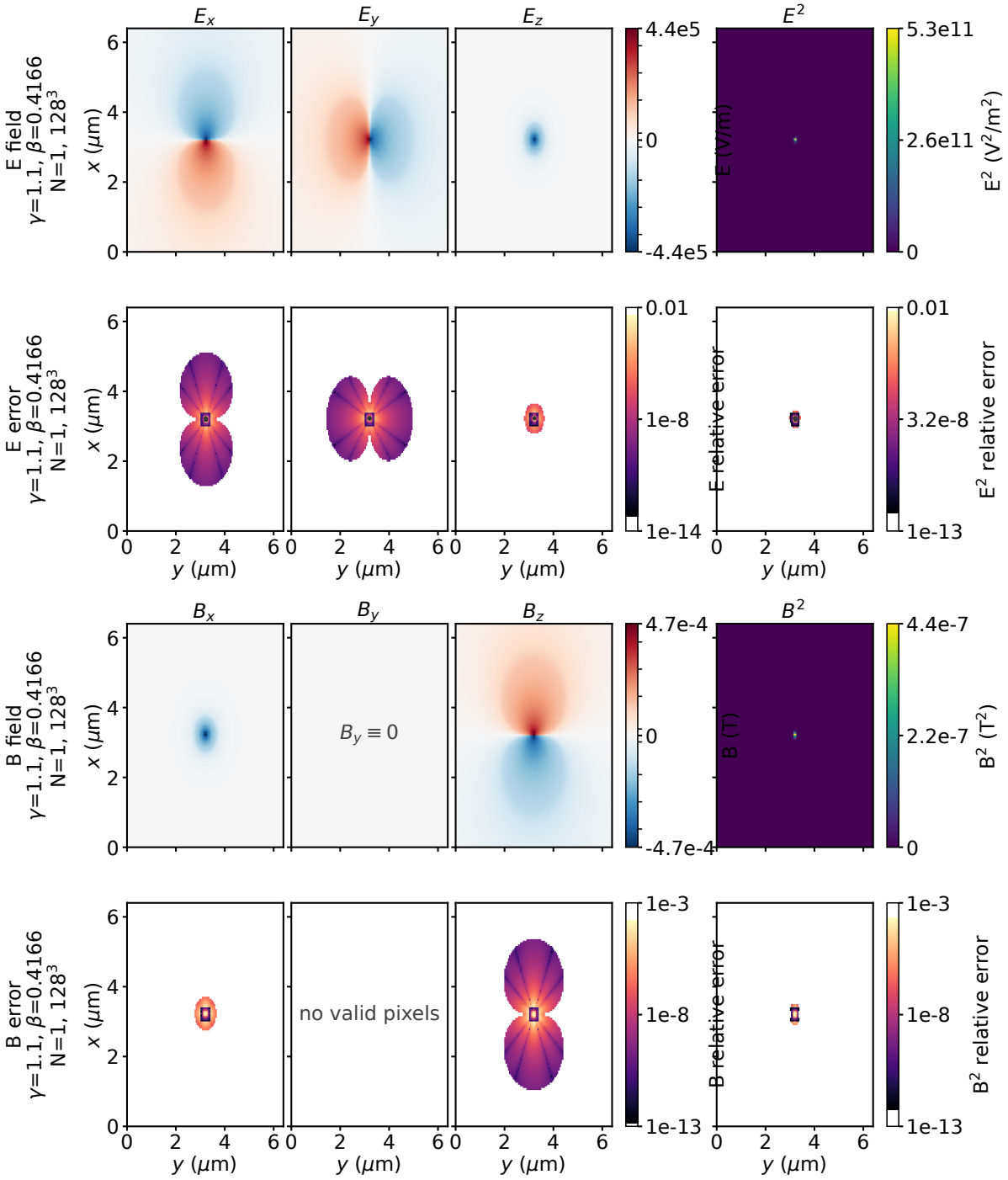


Figure 6: Single-electron field and error on the x-y plane at mid-z for $\beta = 0.4166$ ($\gamma = 1.1$) on the 128^3 grid, cell 5×10^{-8} m, with the same field, colour-bar and error conventions as Fig. 5. The worst component error is 3.669×10^{-3}.

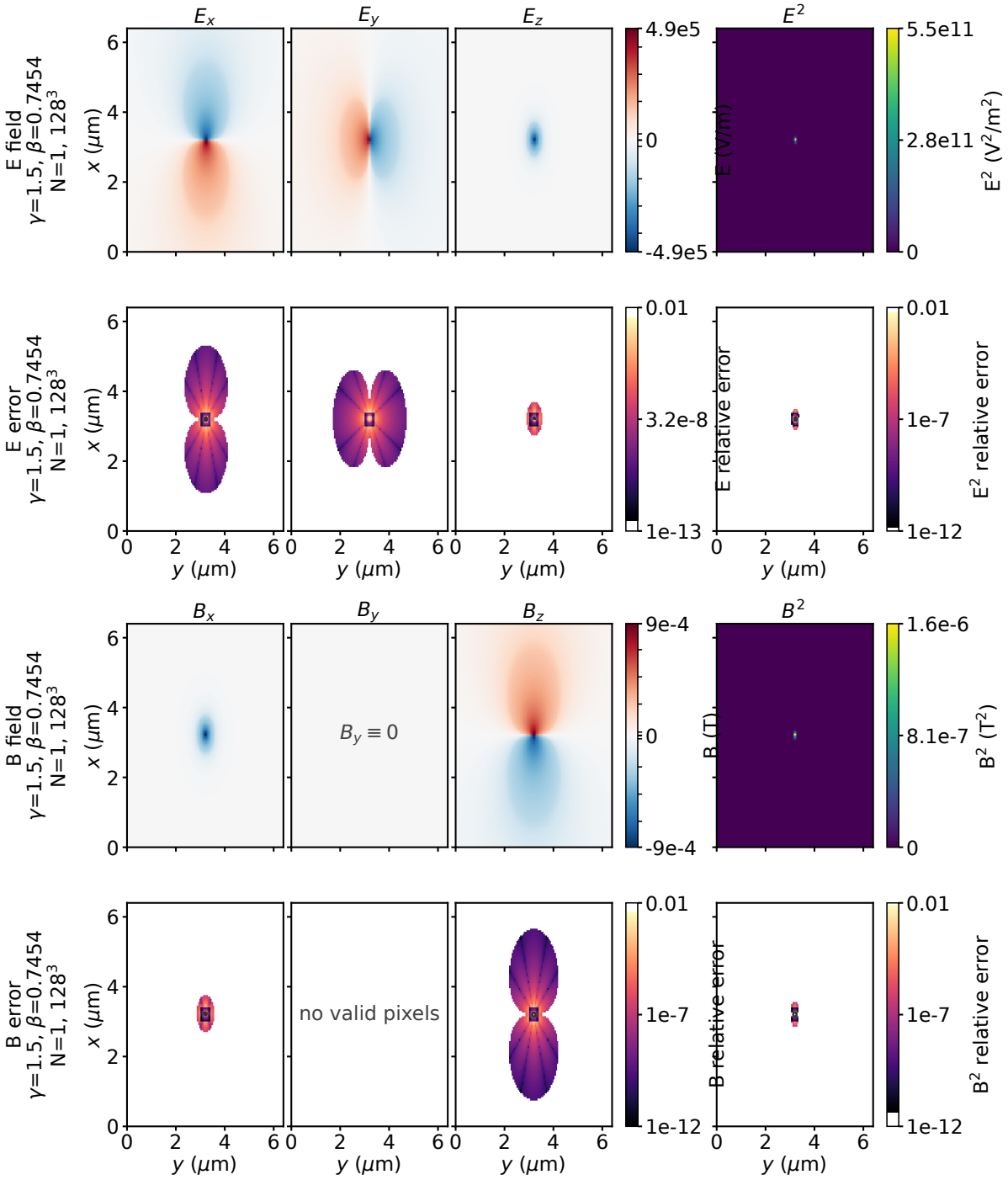


Figure 7: Single-electron field and error on the x-y plane at mid-z for $\beta = 0.7454$ ($\gamma = 1.5$) on the 128^3 grid, cell 5×10^{-8} m, with the same field, colour-bar and error conventions as Fig. 5. The worst component error is 3.719×10^{-3}.

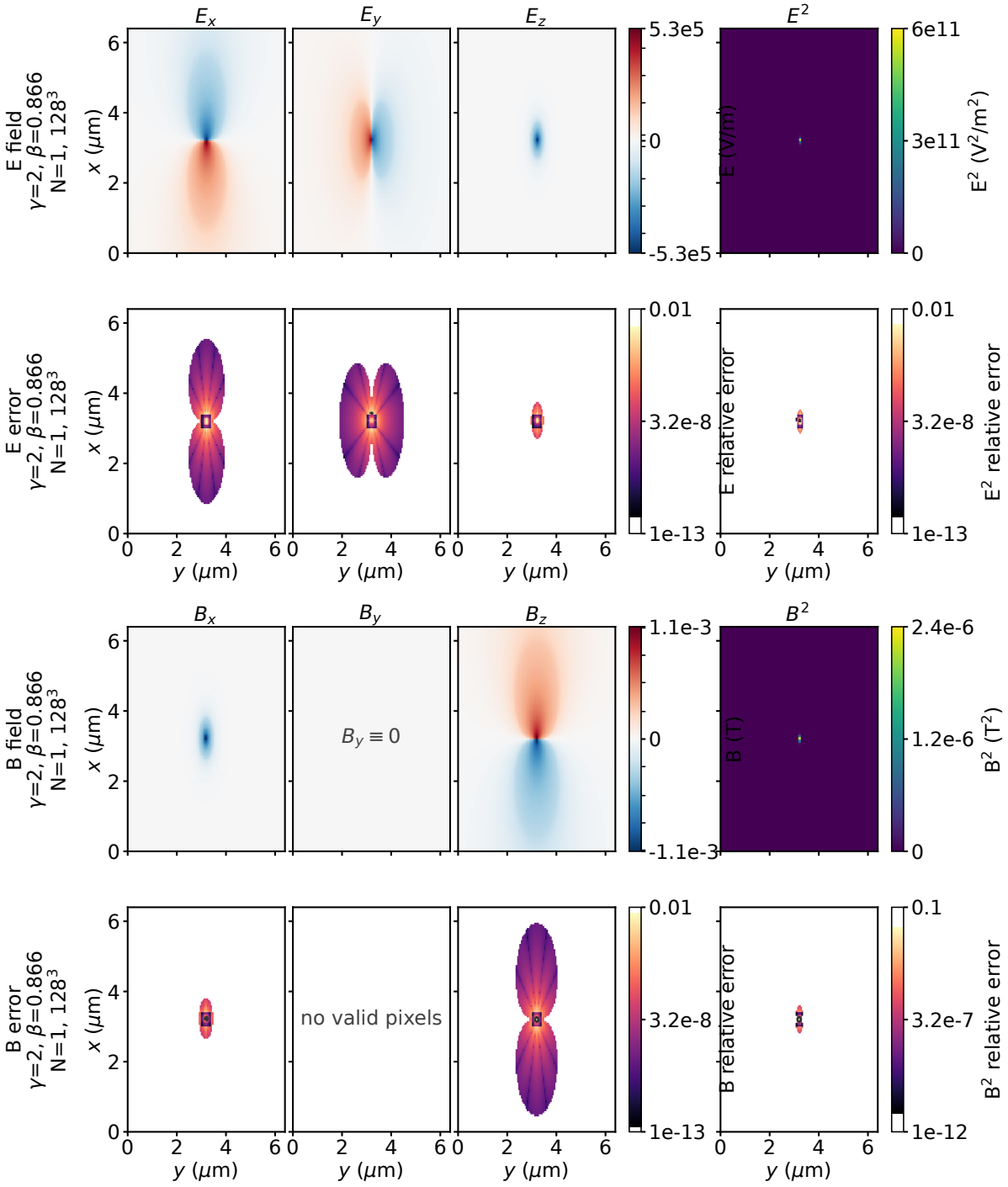


Figure 8: Single-electron field and error on the x-y plane at mid-z for $\beta = 0.8660$ ($\gamma = 2.0$) on the 128^3 grid, cell 5×10^{-8} m, with the same field, colour-bar and error conventions as Fig. 5. The worst component error is 5.098×10^{-3}.

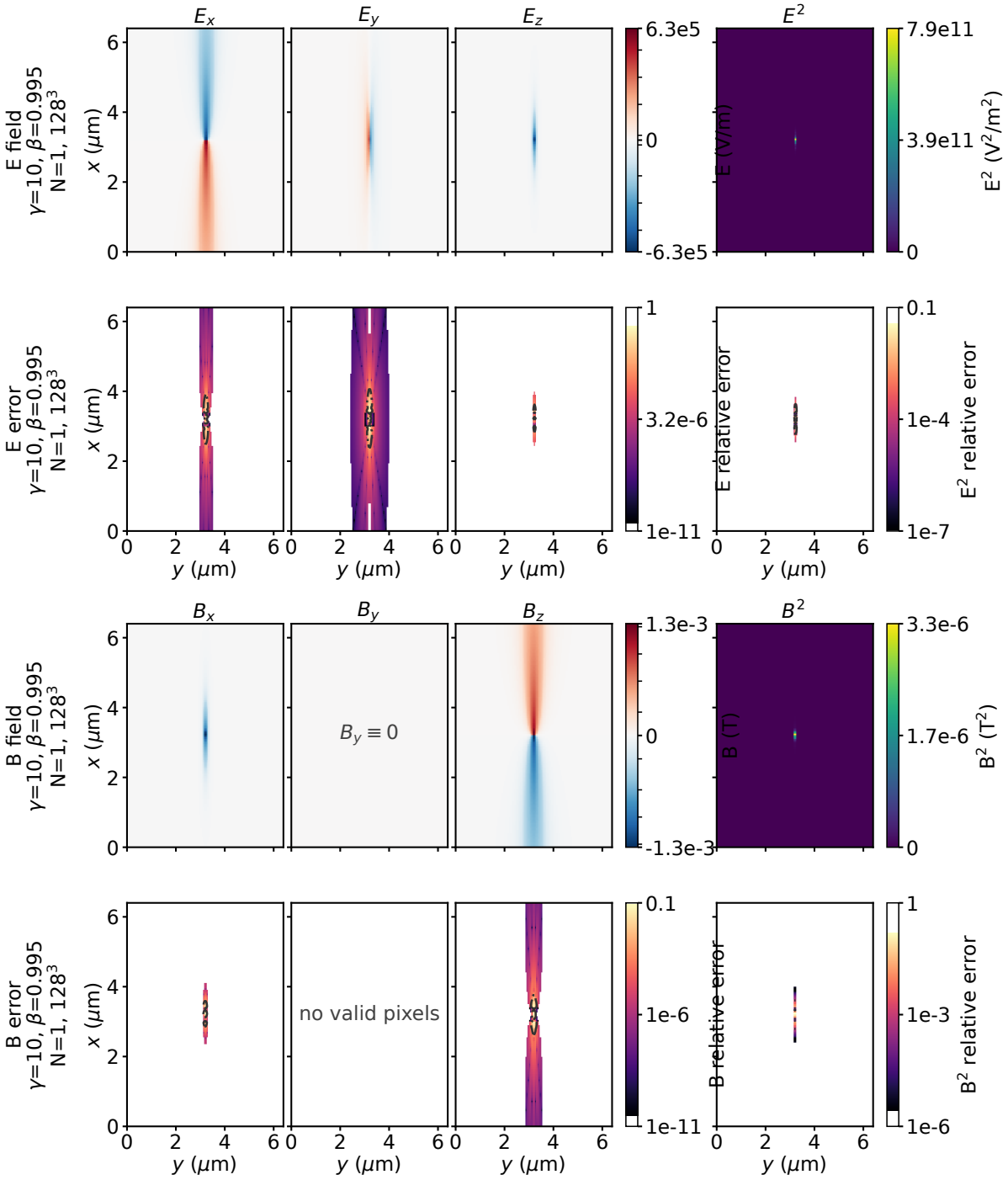


Figure 9: Single-electron field and error on the x-y plane at mid-z for $\beta = 0.9950$ ($\gamma = 10$) on the 128^3 grid, cell 5×10^{-8} m, with the same field, colour-bar and error conventions as Fig. 5. The worst component error is 1.197×10^{-1}.

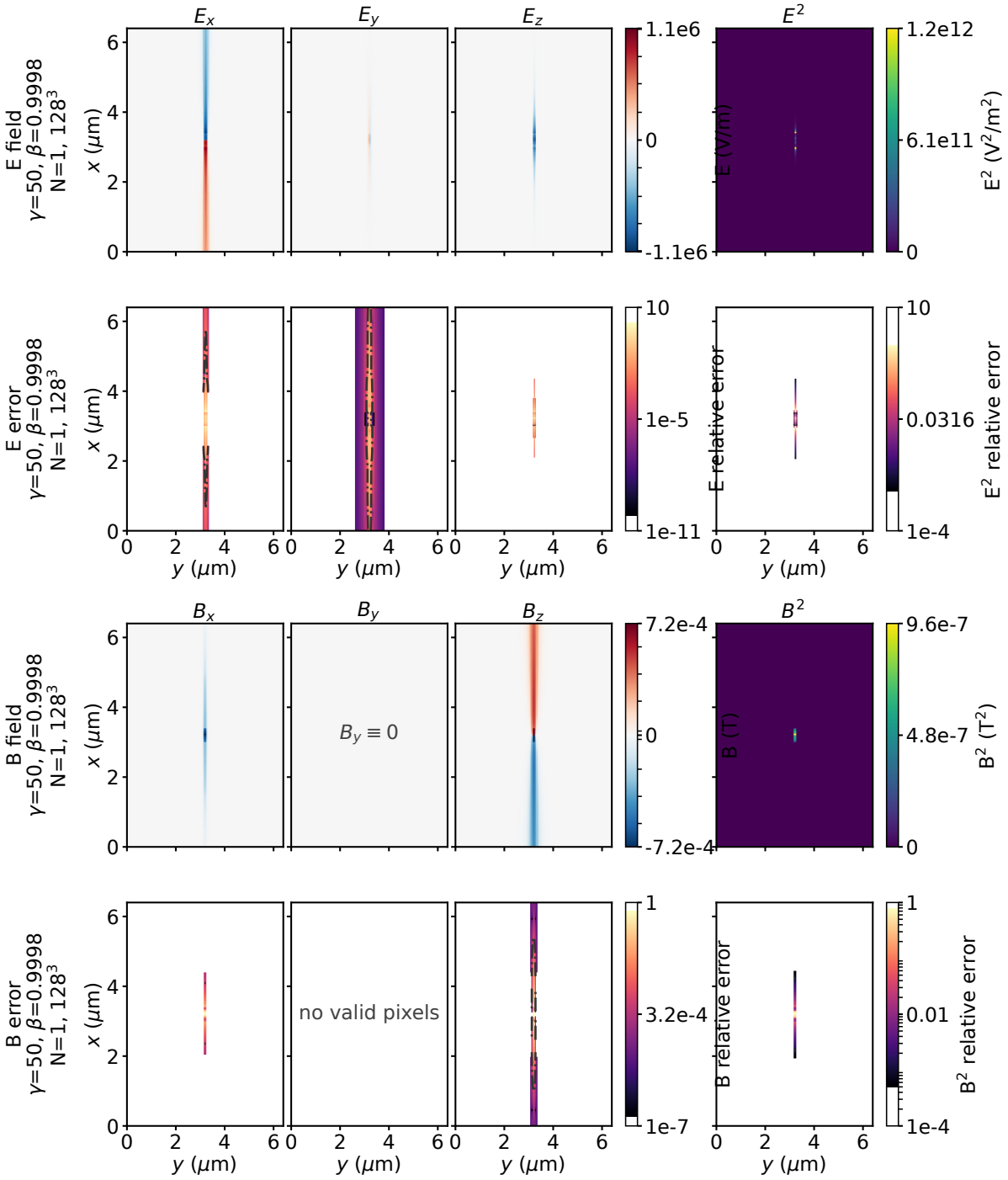


Figure 10: Single-electron field and error on the x-y plane at mid-z for $\beta = 0.99980$ ($\gamma = 50$) on the 128^3 grid, cell 5×10^{-8} m, with the same field, colour-bar and error conventions as Fig. 5. The worst component error is 1.426.

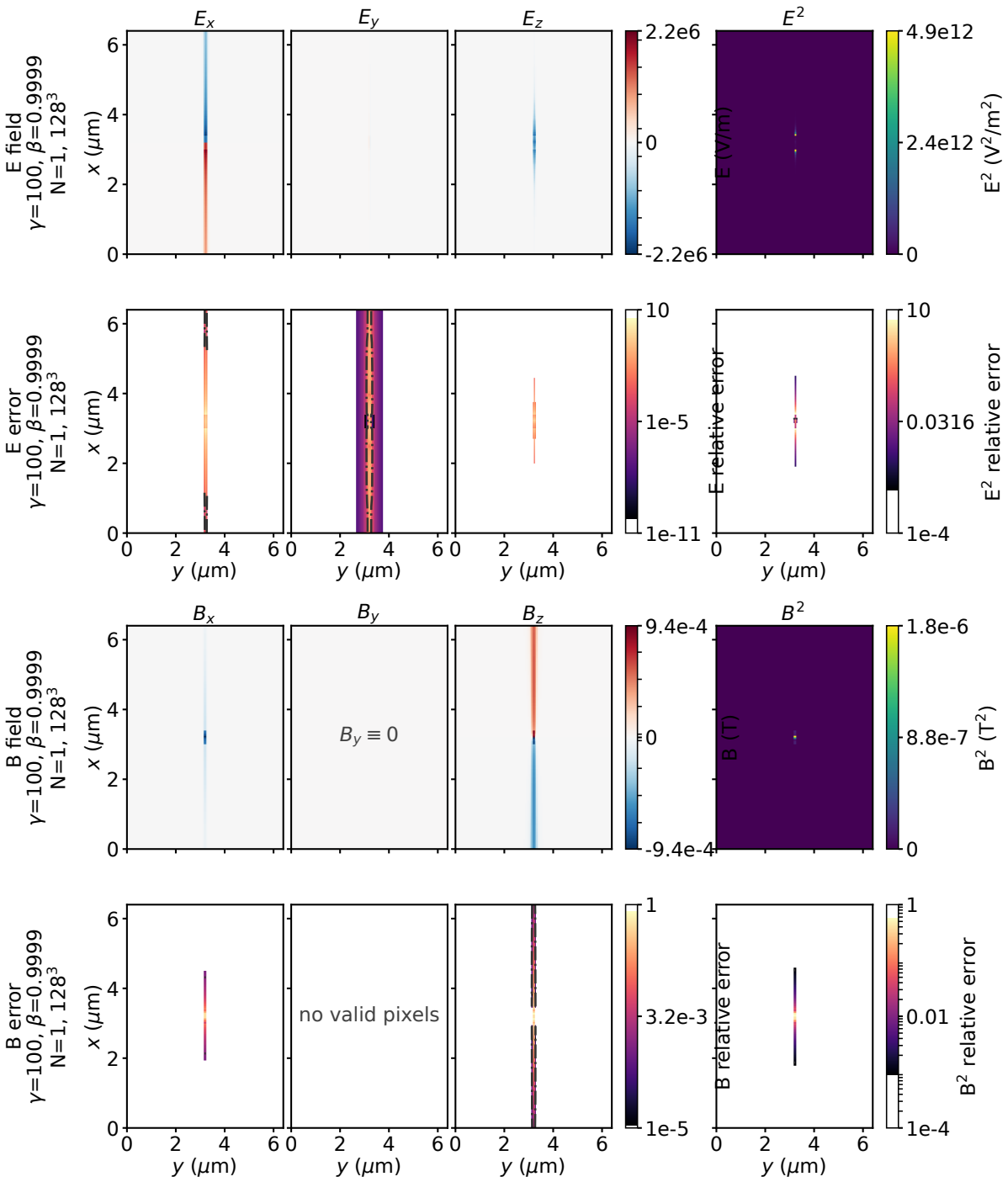


Figure 11: Single-electron field and error on the x-y plane at mid-z for $\beta = 0.99995$ ($\gamma = 100$) on the 128^3 grid, cell 5×10^{-8} m, with the same field, colour-bar and error conventions as Fig. 5. The worst component error is 3.213.

Padding and R2C/C2R details

With pad = 1 the cyclic convolution wraps and a 9.275×10^{-3} periodic-image artefact appears for a single stationary particle in a 20^3 box. For a single stationary particle the pad = 2 result agrees with the direct non-cyclic sum to 3.811×10^{-16} and the pad = 3 result to 1.905×10^{-16}; at pad ≥ 2 the error is at machine precision ($\leq 10^{-15}$). The artefact grows for beamed sources: at pad = 1 the 100-source $\gamma = 10$ cloud shows a 7.5×10^{-1} worst-component error.

For the R2C/C2R half-spectra the DC bin and the seven self-conjugate even Nyquist corners must

be exactly real; the intermediate Nyquist planes are Hermitian-paired, not real. The R2C/C2R path matches the C2C path to 1.665×10^{-16} over all six components.

The γ-landscape runs of Section 5.5 used padding 1 (padding 2 was infeasible for the 384^3 $h = 1/4$ grid); the $d = 48$ rows carry a small padding-1 periodic-wraparound floor of order 10^{-4} to 10^{-3}, so the $d = 48$ values are not padding-clean.

C Mid-bunch slices and large-domain slices

C.1 Mid-bunch slices: field and error side by side

The distance sweeps of Section 5.4 sample the far field. To measure the mesh in the bunch core we slice through it and compare, pixel by pixel, with an exact TSC-convolved reference: 64×64 planes, cell 10^{-7} m, $N = 200$, $Q = -100$ pC, $\gamma = 10$ with momentum along $+y$. Two cases share the positions and differ only in a 1.045 mrad rms transverse divergence in x (Figs. 12, 13, 14, 15); a separate 32×32 plane carries the binned 1 mrad case (Fig. 16). In every panel y is horizontal and increases left to right, so the bunch moves rightwards and y is the beam axis; the vertical axis is x in the x-y cut and z in the z-y cut. Each figure has a field row with four columns (E_x, E_y, E_z, E^2) and an error row underneath. The three components share one linear colour scale and bar; E^2 (or B^2) has its own scale and bar with its own unit label. The error row is logarithmic and masked where the reference is below 10^{-3} of its own peak. The forward case compares the mesh plus fast shell against the order-4 adaptive Gauss–Legendre driver of the original campaign. That reference is now superseded by the Lorentz-contracted in-support quadrature (Section 5.6); the reimplementation changes the closest-node reference at the few-percent level, so the in-support error pixels in these panels are an upper bound rather than a current absolute error. The solid white outline marks the TSC support. The quantitative in-support statement remains the fast-versus-arbiter 1.6% at the worst node (Section 5.6). The fluctuation panels show the complete reference field; the implementation carries the binned momentum-spread case through the multi-velocity path (Appendix D).

Shell radius unit. The near-shell envelope measured in the crossover work, "3 to 4 PIC cells," assumed a mesh spacing $h = 1/4$ of the PIC cell, so $R = 3$ and $R = 4$ are 12 and 16 mesh nodes. On this grid the node spacing is $h = 1$ PIC cell (2×10^{-7} m), so the same $R = 3/4$ is $3/4$ mesh nodes, four times smaller in physical units than the envelope configuration. At $R = 3$ on an $h = 1$ grid the shell covers 3 mesh nodes; the TSC support extends ± 1.5 cells from each source, so the shell covers a fraction of the full support. The measured plane errors at this radius are reported below.

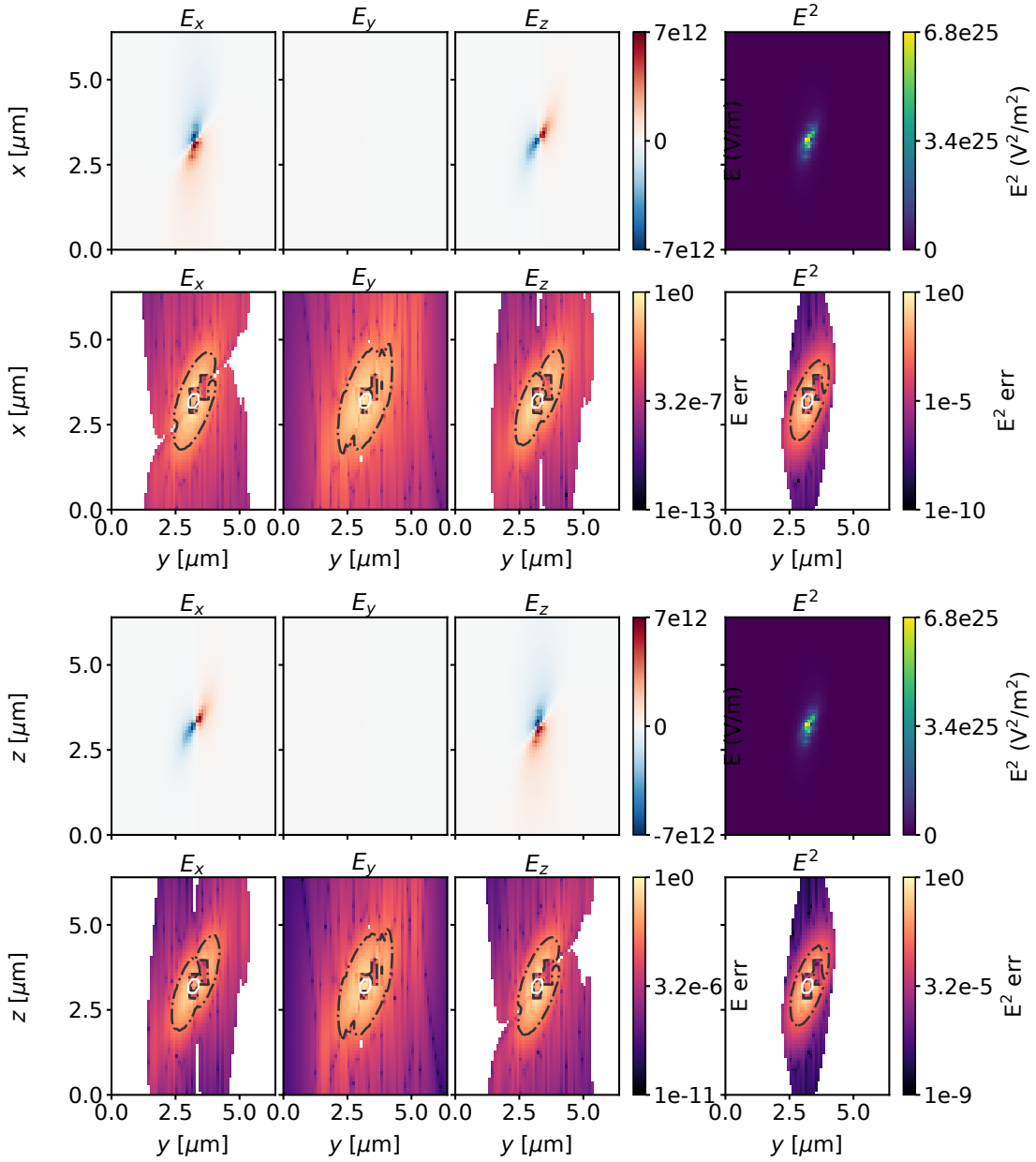


Figure 12: Forward-case electric validation ($N = 200$, $Q = -100\,\mathrm{pC}$, $\gamma = 10$ along $+y$, 64×64 planes, cell 10^{-7} m; x-y plane at $z = 3.2\,\mu\mathrm{m}$ (upper) and z-y plane at $x = 3.2\,\mu\mathrm{m}$ (lower), with y horizontal, the beam axis. Each image shows four columns E_x, E_y, E_z, E^2: the three components share one linear scale and bar, while E^2 has its own scale and unit label; the lower row is logarithmic error, masked below 10^{-3} of the peak, and the white outline marks the TSC support. The in-support fast-versus-arbiter error against the order-12 independent continuum reference is 1.6% at the worst node.

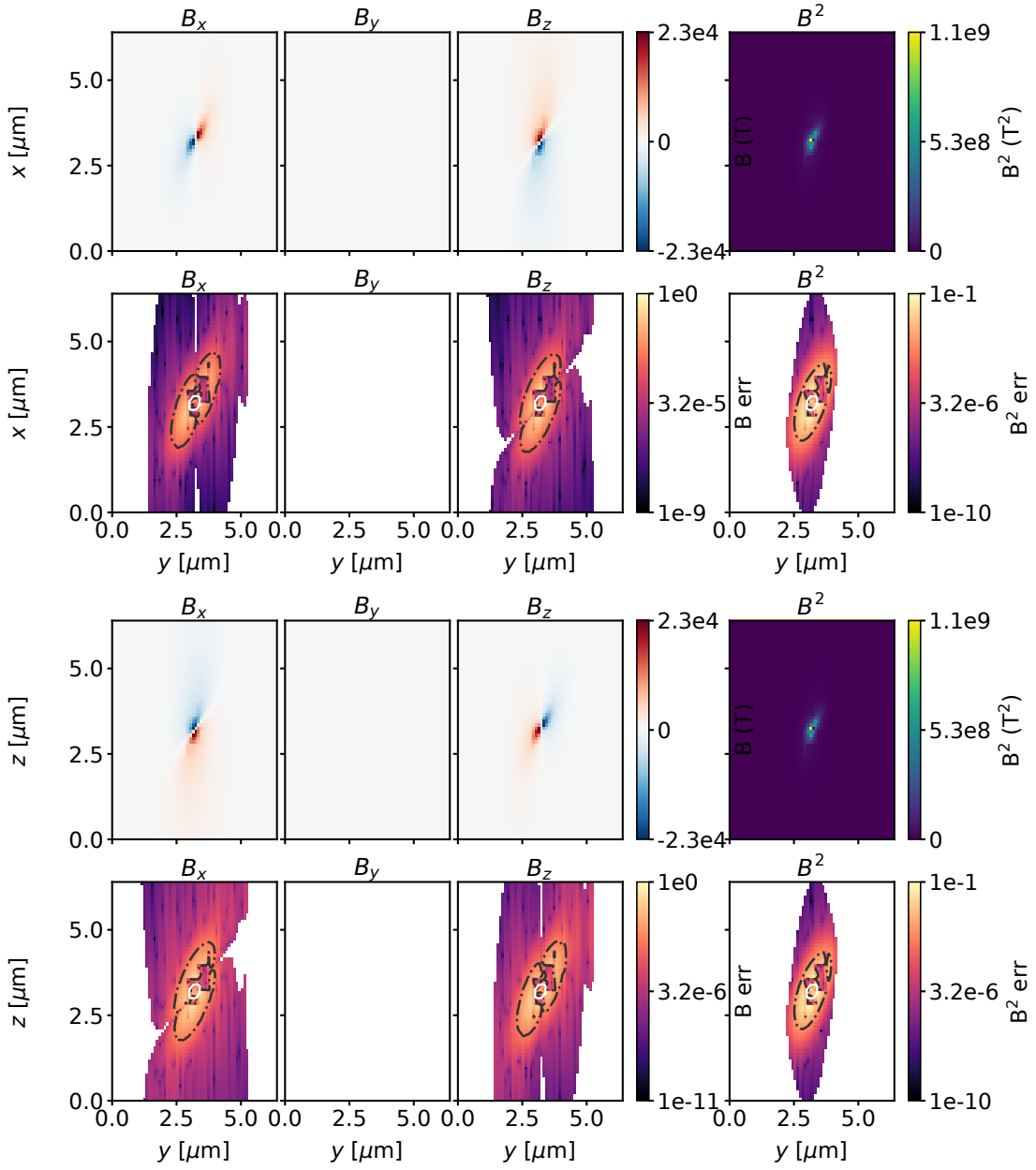


Figure 13: Magnetic companion to Fig. 12 with the same planes and layout, showing four columns B_x, B_y, B_z, B^2: the three components share one linear bar and B^2 has its own scale and unit label; $B_y \equiv 0$ for motion along $+y$ is marked in its panel. y is horizontal and is the beam axis, and the same TSC-support outline is drawn as in the electric figure. The magnetic field is about 10^{-10} of the E-dominated local field at $\gamma = 10$.

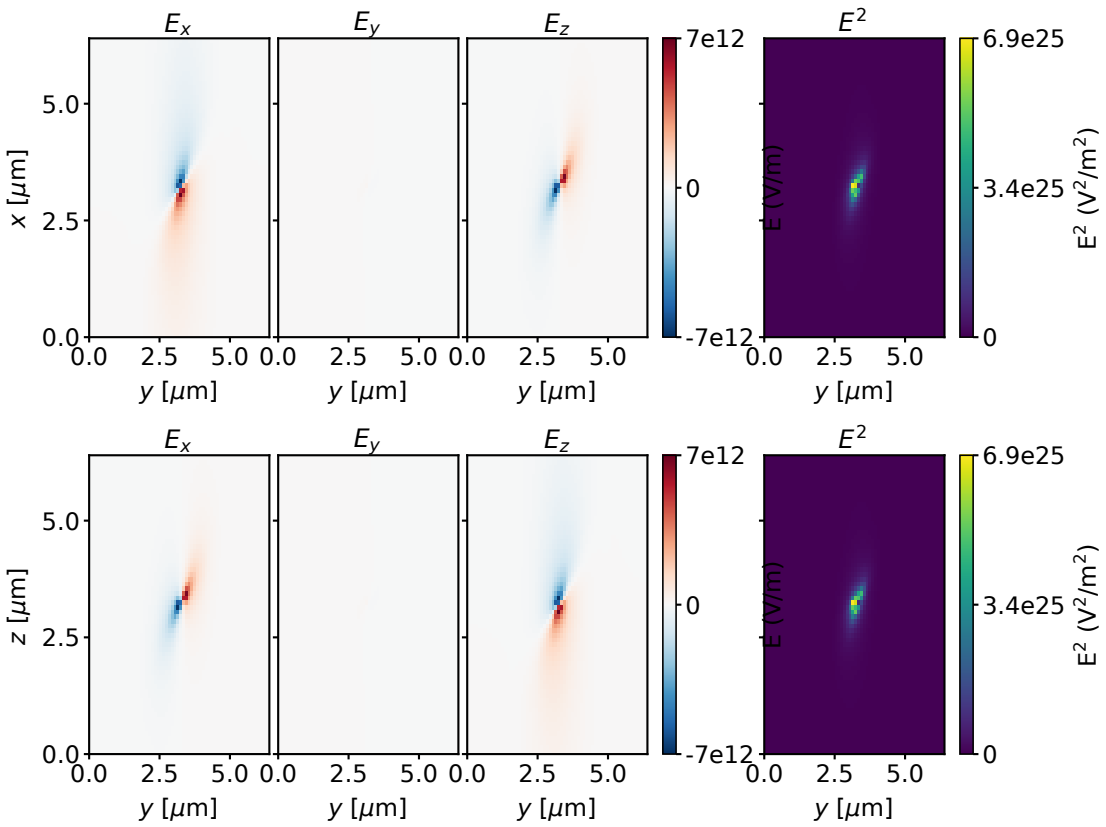


Figure 14: Fluctuation-case electric reference with the same geometry and bunch as Fig. 12 but a 1.045 mrad rms x-divergence, showing four columns E_x, E_y, E_z, E^2 from the order-12 independent continuum reference: the three components share one linear scale and bar, while E^2 has its own scale and unit label. y is horizontal and is the beam axis. The reference field is shown; the corresponding binned picBunch panels were not part of this campaign.

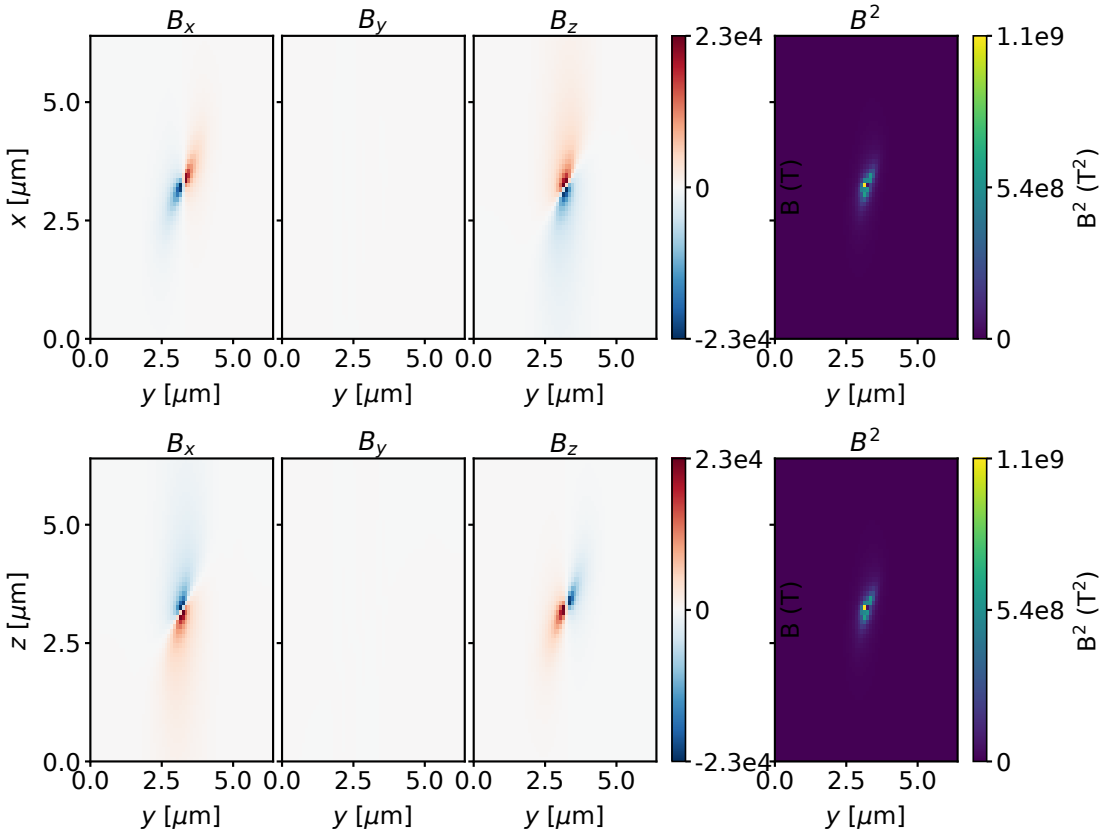


Figure 15: Magnetic reference for the fluctuation case, with the same planes, four columns B_x, B_y, B_z, B^2: the three components share one linear bar and B^2 has its own scale and unit label, as in Fig. 14; y is horizontal and is the beam axis. The x-divergence lifts B_y off zero, although it remains about 3×10^{-4} of the dominant component. Only reference fields from the order-12 independent continuum reference are shown; the picBunch panels are unavailable.

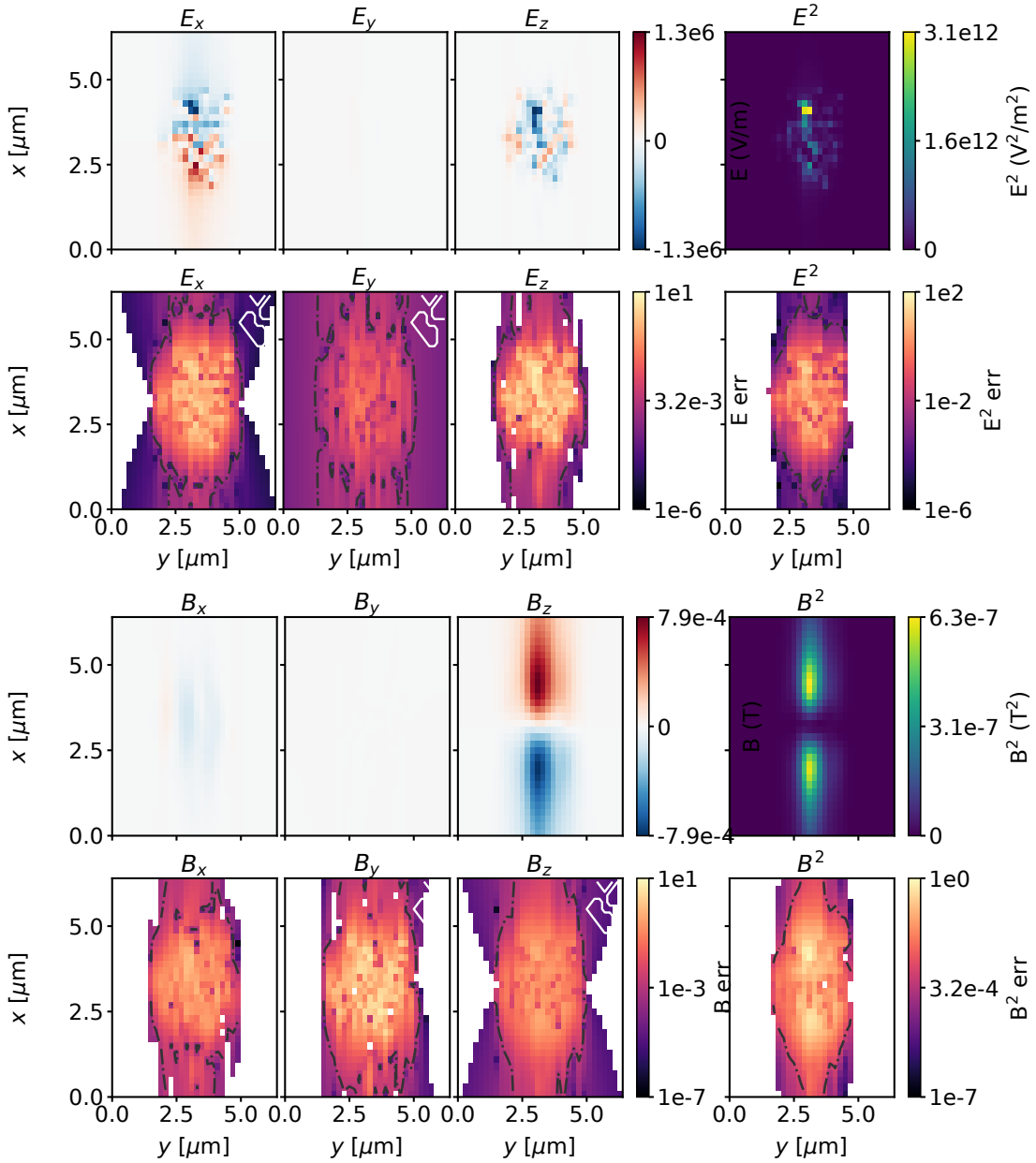


Figure 16: Binned 1 mrad case ($N = 200$, $Q = -32$ aC, $\gamma = 10$, 32×32 x-y plane at $z = 1.6$ µm, cell 2×10^{-7} m), with the picBunch velocity-binned field against the order-12 independent continuum reference. The upper image is electric and the lower magnetic, each with four columns: three components on one shared scale and bar, and E^2 or B^2 on its own scale and bar; y is horizontal and is the beam axis, and error rows are logarithmic and masked below 10^{-3} of the reference peak. The solid white outline marks the TSC support.

The error is largest where these planes cut the bunch core: the spectral far-field convolution is not the dominant term there, and the maps show the measured error concentrated at the beamed core and falling away toward the plane edges. The far field is measured separately (Table 3, Fig. 1); the near-shell contribution is measured at the same geometry below.

C.2 Near-shell accuracy and cost on the slice geometry

On the same $N = 1000$, $32 \times 32 \times 16$, cell 2×10^{-7} m geometry, the exact near shell was run through the library path at the radius the envelope case assumed, $R = 3$ for $\gamma = 1$ and $R = 4$ for $\gamma = 10$, in mesh nodes of this $h = 1$ grid. The mesh-only plane errors (Table 17) are 0.12–0.415 for $\gamma = 1$ and 0.11–4.60 for $\gamma = 10$: these planes cut the sources, the region where the spectral convolution is not the

dominant term. With the near shell the worst per-component plane error is 1.6×10^{-3}–1.1×10^{-1} for $\gamma = 1$ (the high end is the z-y plane E_z, 0.1067) and 3.2×10^{-2}–5.7×10^{-1} for $\gamma = 10$: at $R = 3/4$ on an $h = 1$ grid the shell radius covers $3/4$ mesh nodes, which is a fraction of the TSC support extent, and the measured residual is the mesh contribution outside the shell. The discrete Gauss-law indicator, the Yee backward-difference div E residual normalised to the peak deposited ρ/ε_0, moves from 0.3534 to 0.0895 for $\gamma = 1$ and from 5.026 to 0.399 for $\gamma = 10$, against reference values 0.0904 and 0.0814: $\gamma = 1$ reaches the reference value, $\gamma = 10$ drops 12.6× and stays 4.9× above it.

The assembly itself is not the limiting term: the tests of Section 3.6 show the tiled subtraction exact to roundoff, so the residuals above are the mesh outside the shell. Runtime at $N = 1000$ is 4.3 min for $\gamma = 1$ and 59 min for $\gamma = 10$ (one 16-thread socket); the exact evaluator costs about 8× more per pair at $\gamma = 10$ than at $\gamma = 1$.

C.3 Large-domain slices

Figure 17 shows the $\gamma = 10$ field and error on a $128 \times 128 \times 64$ domain, cell 2×10^{-7} m, $N = 100$, against the independent continuum at tolerance 10^{-3}; Fig. 18 is its magnetic companion, and Fig. 19 is the stationary-bunch electric case. The boundary field is about 10.3% of the peak (worst per-component, $\gamma = 10$ z-y E_z), so the colour scale is set by the bunch and the far decay is visible inside the frame. The core error is the method error of the mesh where the planes cut the bunch, far above the 10^{-3} reference floor.

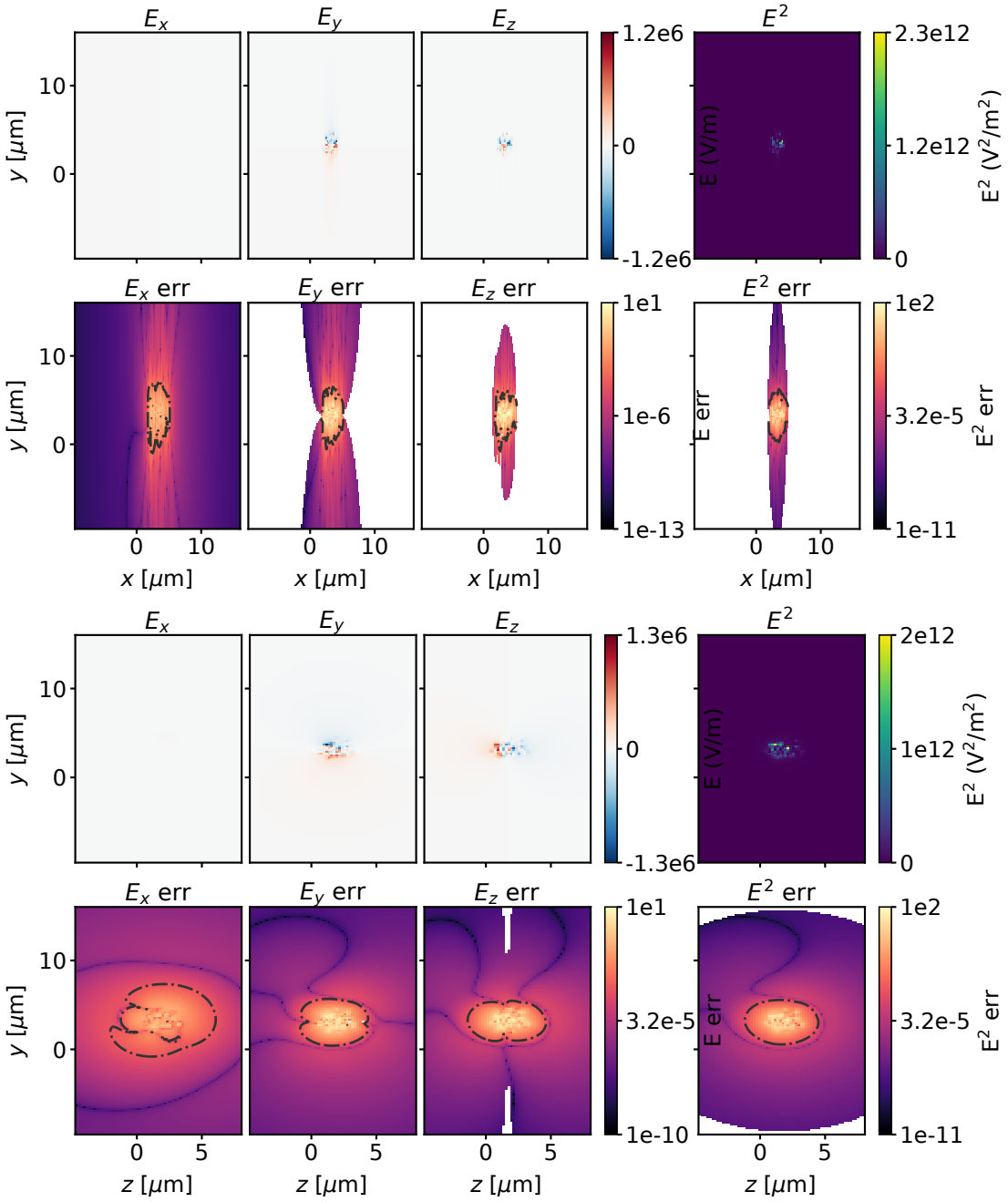


Figure 17: Large-domain electric slices ($\gamma = 10$, $128 \times 128 \times 64$, cell 2×10^{-7} m, $N = 100$, `jrc0880`) from the earlier axial-x data set (not the $+y$ beam-axis set); x is horizontal in the x-y cut and is the beam axis there, while z is horizontal in the z-y cut. Upper and lower images are x-y and z-y planes, each showing four columns E_x, E_y, E_z, E^2: the three components share one linear scale and bar, and E^2 has its own scale and unit label, above a logarithmic error row against the order-4 driver reference. The boundary field is 10.3% of the peak (worst component, z-y E_z), so the colour scale is set by the bunch.

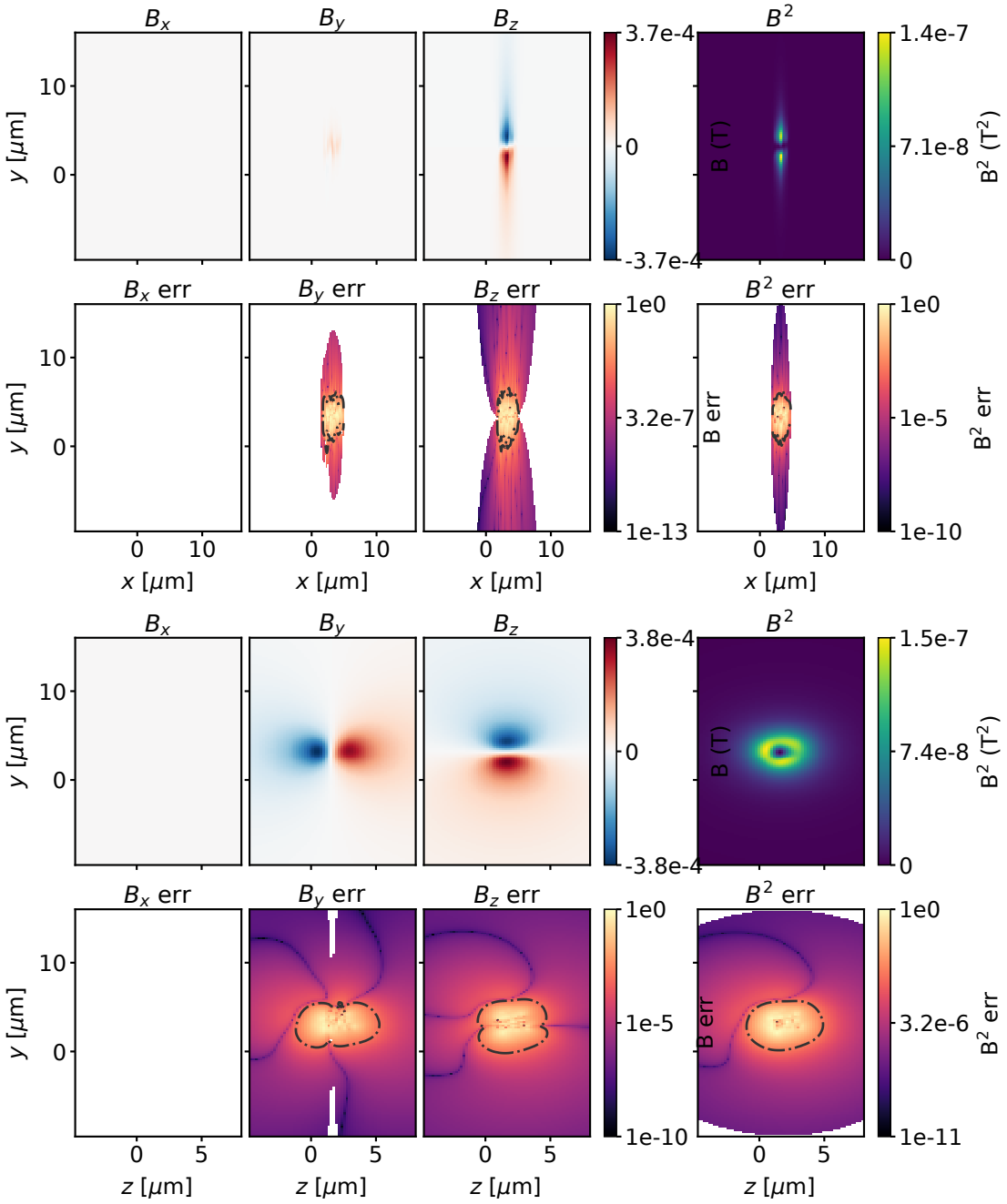


Figure 18: Magnetic companion to Fig. 17 at $\gamma = 10$, same planes and earlier axial-x geometry; x is horizontal in the x-y cut and is the beam axis there, while z is horizontal in the z-y cut, and $B_x \equiv 0$ is marked in its panel. Each image shows four columns B_x, B_y, B_z, B^2: the three components share one linear bar and B^2 has its own scale and unit label, with the same logarithmic error convention as the electric figure. The plane peak is 6.2×10^{-4} T.

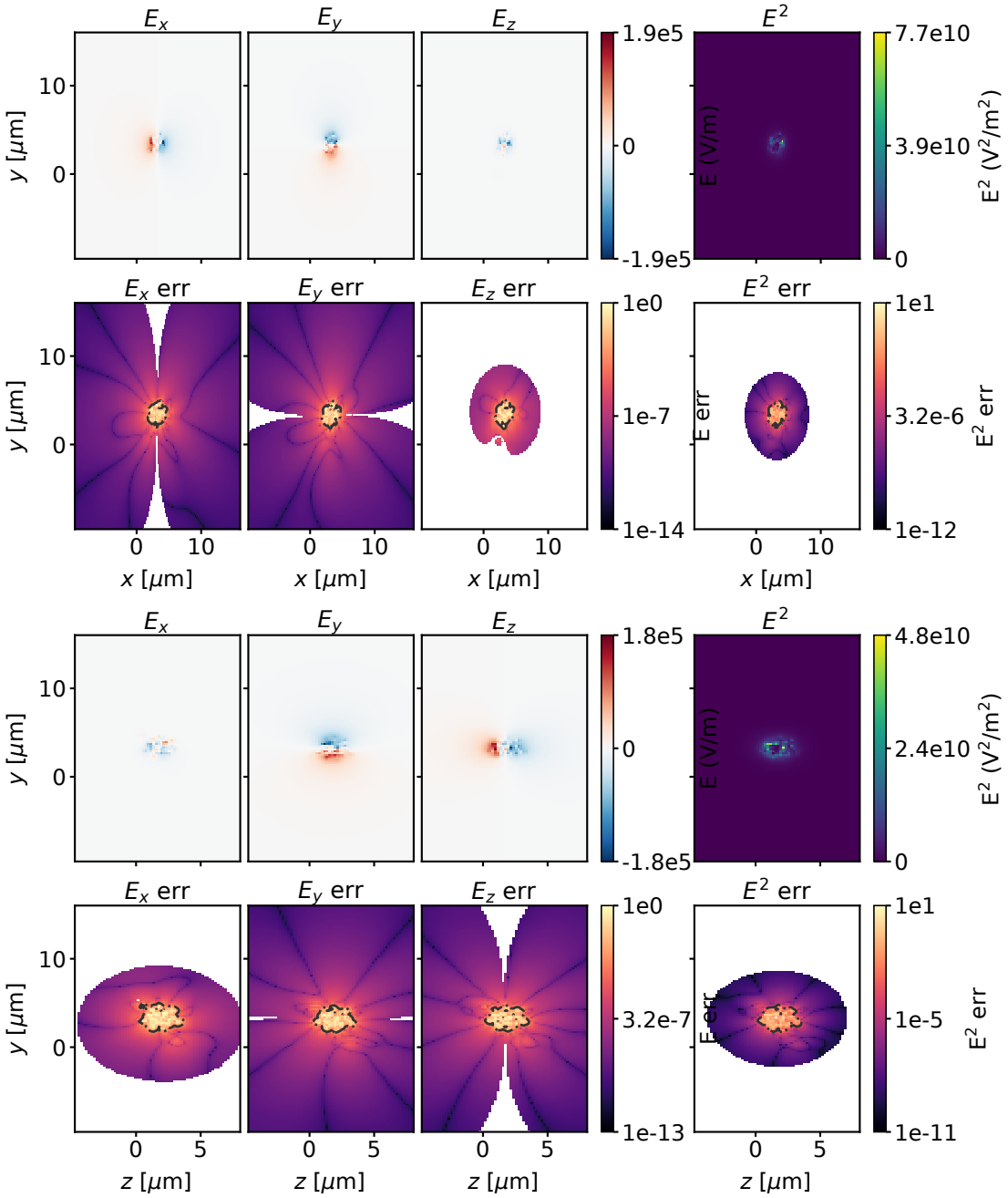


Figure 19: Large-domain electric slices for the stationary bunch ($\gamma = 1$), same domain and layout as Fig. 17; x is horizontal in the x-y cut and z is horizontal in the z-y cut (the bunch is stationary, so there is no beam axis). Each image shows four columns E_x, E_y, E_z, E^2: the three components share one linear bar and E^2 has its own scale and unit label, above a logarithmic error row. The core error reaches 7.6×10^{-1} (z-y E_x), far above the 10^{-3} reference floor.

D Multi-velocity binning, measured

D.1 Binned convolution

The single-velocity scope of Section 2 is extended by splitting the velocity distribution into B classes and running one convolution per class; the near shell needs no binning, because each pair carries its own velocity. The measurements below are on a 64^3 grid with 200 sources on `hal` (NVIDIA A30, `sm_80`, CUDA 13.4.92).

Box representative. Each box is represented by its absolute-charge-weighted mean momentum, not by the geometric box centre. The mean minimises the angular and energy deviations for which the error model charges the split, at no extra kernel cost. On a 200-particle, $\gamma = 10$, 5° bunch the global L2 field error against the 128-box field falls from 13.2% to 11.2% at 16 boxes and from 4.31% to 4.13% at 64 boxes. The raw rows are in `docs/measurements/velocity_bin_representative.csv`.

Single-velocity sanity. With zero spread the binned path is bitwise equal to the unbinned path: $B = 1$ and $B = 16$ give bit-identical fields in all six components ($\max|\Delta| = 0$).

Figure 20 shows the sampled distributions and the bin comparison. With a 5% energy spread and $+y$ drift the B_z peak is 2.32050×10^4 T at 16 bins and 2.32049×10^4 T at 64 bins, 0.0003% apart; $B_y = 0$ exactly, which is correct for a purely longitudinal current. With a 5° angular spread the B_z peak moves from 2.48202×10^4 to 2.48213×10^4 T and B_y from 1294.8 to 1295.5 T; the ratio B_y/B_z is 5.22×10^{-2}, and the two bin counts agree to 0.0043% in B_z and 0.056% in B_y.

1 mrad case against the arbiter. The fluctuation bunch ($N = 200$, $\gamma = 10$, $+y$ drift, 1.046 mrad rms x-divergence) was re-run at 64 bins and compared against the Gauss–Legendre arbiter (unbinned, per-particle u, 698-node slice subset) as the reference. On the x-y slice ($z = 32$) picBunch gives $B_y = 7.835$ T and $B_z = 2.3336 \times 10^4$ T, against arbiter values of 7.759 T and 2.344×10^4 T (factors 1.010× and 0.996×); on the z-y slice ($x = 32$) the corresponding values are 7.561 T / 7.519 T and 2.3159×10^4 T / 2.327×10^4 T (1.006× and 0.995×). The 3-D B_y peak is 8.734 T and B_z is 2.3336×10^4 T. The worst component deviation is 1.4% (x-y ratio); every component on every slice is within 1.0%.

Runtime: $B = 64$, 64^3, 200 particles, `-near-shell fast` ($R = 3$, tile 4, order 5), 248 s on the A30.

Far-tail switch validation

For the documented 20-cell, 4096-electron bunch (source radius $R = 16.85$ cells, peak 45722.5 V/m, tolerance level 45.72 V/m), the quadrupole margin of Eq. (11) is 350 cells, which exceeds the 2^{28} padded-node cap, so the code falls back to the measured far-tail switch. Table 19 gives the switch error: the worst per-component $|\mathbf{E}_{\text{tail}} - \mathbf{E}_{\text{direct}}|$ against an independent direct charge sum in 13 directions.

A 512^3 output sampled outside the compute box confirms the switch in practice: 36 nodes including corners, worst $|\mathbf{E}_{\text{file}} - \mathbf{E}_{\text{direct}}| = 2.8 \times 10^{-3}$ V/m against the 45.7 V/m tolerance level (6.1×10^{-5} of it), and every sample nonzero, i.e. no interior zero fill.

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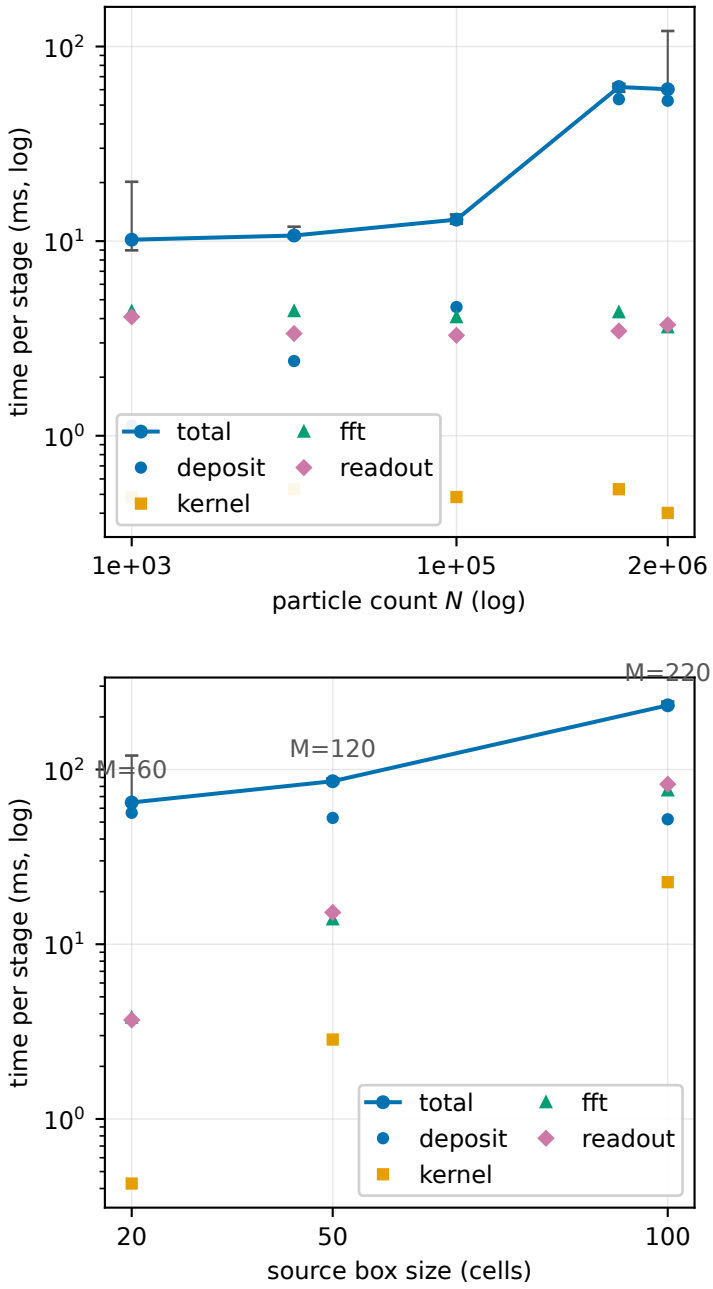


Figure 3: Pipeline time versus particle count (top, log-log) and source-box size (bottom, semi-log), padded array $M = 60/120/220$ cells, $\gamma = 10$, $h = 1$, double precision, `rosi_v100`/CUDA 12.6. Blue is the total and markers the per-stage medians; the deposit dominates at $N = 2 \times 10^6$ (57 of 64.8 ms), and the total rises from 64.8 to 233 ms as M grows from 60 to 220.

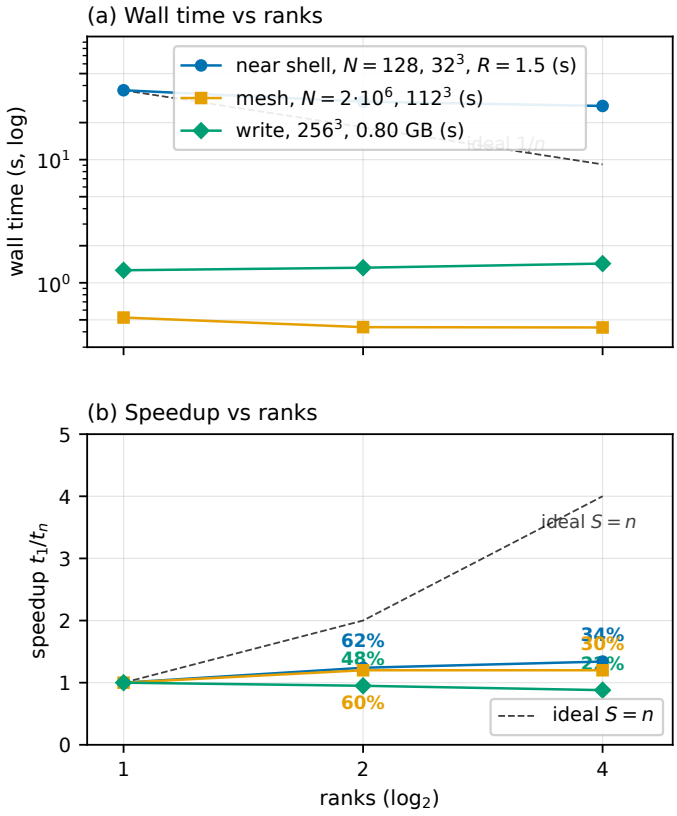


Figure 4: Strong scaling of the three rank-parallel pieces on `rosi_v100` (idle window, medians of three runs) for 1–4 ranks, with dashed ideal $1/n$ and $S = n$ references. The near shell reaches 1.34×, the component-parallel mesh 1.20×, and the collective writer stays within 13% from one to four ranks.

Table 13: Configurations of every measured case and the section that uses it (recorded values of this work; "not recorded" means no value was recorded). The two mid-slice entries give the bunch generation grid; the field grid for those cases ($32 \times 32 \times 16$ @ 2×10^{-7} m) is in the second block. Kinetic energy is $(\gamma - 1)m_e c^2$; the multi-velocity driver cell size is 10^{-6} m.

case	N	total charge ($-e$ each)	size (cells/axis)	spread	used in
mid slices, g10	1000	1.602×10^{-16} C	bunch generation grid $64{\times}64{\times}32$ @ 10^{-7} m ($6.4 \times 6.4 \times 3.2$), $\sigma{=}6$ cells/axis	none ($\gamma_{\min} = \gamma_{\max} =$ 10)	§C, §6
mid slices, g1	1000	1.602×10^{-16} C	bunch generation grid $64{\times}64{\times}32$ @ 10^{-7} m ($6.4 \times 6.4 \times 3.2$), $\sigma{=}6$ cells/axis	none ($\gamma_{\min} = \gamma_{\max} =$ 1)	§C, §6
large-domain, g10	100	1.602×10^{-17} C	extent $2.74 \times 3.78 \times 2.65$	not recorded	§C (Fig. 17)
large-domain, g1	100	1.602×10^{-17} C	extent $2.74 \times 3.78 \times 2.65$	not recorded	§C (Fig. 19)
cost, stationary	2×10^6	not recorded	20-cell bunch	none	§5.7
cost, moving axial	2×10^6	not recorded	20-cell bunch	none	§5.7
landscape, aspect 1	1	1.602×10^{-19} C	single particle, 96-cell box	none	§5.5
landscape, aspect 4	1	1.602×10^{-19} C	single particle at the origin	none	Table 2
multi-velocity, E 5%	200	3.2×10^{-17} C	$\sigma = 6$ cells, centroid $(32, 32, 32)$	5% energy, $+y$ drift	§D
multi-velocity, A 5°	200	3.2×10^{-17} C	$\sigma = 6$ cells, centroid $(32, 32, 32)$	5° angular, $+y$ drift	§D
1 mrad, binned	200	3.2×10^{-17} C	$\sigma = 6$ cells, centroid $(32, 32, 32)$, 1.046 mrad rms x-div	1 mrad, $+y$ drift	§D
divergence, 1 mrad	1000	1.602×10^{-16} C	$\sigma = 0.6$ each axis, centroid $(3.197, 3.208, 1.615)$	1 mrad paraxial, $+y$ (seed 43)	§C (reference pending)

case	γ, kinetic E ($m_e c^2 =$ 510.999 keV)	weight, cell	field grid (domain), host
mid slices, both	10 / 5.06 MeV; 1 / 0	1 each; 2×10^{-7} m	$32 \times 32 \times 16$ ($6.4 \times 6.4 \times 3.2$), rosi_v100
large-domain, both	10 / 5.06 MeV; 1 / 0	not recorded; 2×10^{-7} m	$128 \times 128 \times 64$ ($25.6 \times 25.6 \times 12.8$), jureca-h100
cost, both	1 / 0; 10 / 5.06 MeV	not recorded; 2.5×10^{-7} m	$66 \times 68 \times 66$, jureca-h100
landscape, aspect 1	10/50/100	1; 10^{-7} m	96^3 ($h{=}1$), 384^3 ($h{=}1/4$), jureca-h100 (ref: rosi_v100)
landscape, aspect 4	1/10/100	1; anisotropic, aspect 4	axis-aligned, $h = 1$, host not recorded
multi-velocity, both	10 / 5.06 MeV	1 each; 10^{-7} m	64^3, hal (A30, `sm_80`)
divergence	10 / 5.06 MeV	1 each; 2×10^{-7} m	$32 \times 32 \times 16$ ($6.4 \times 6.4 \times 3.2$), rosi_v100

Table 16: Seven-velocity single-particle validation on the final 128^3 grid, cell 5×10^{-8} m, 6.4 μm domain, against the order-12 continuum reference. Worst is the maximum relative error over the six components on the mid-z x-y plane.

γ	β	worst relative error
1.0	0	3.474×10^{-3}
1.1	0.4166	3.669×10^{-3}
1.5	0.7454	3.719×10^{-3}
2.0	0.8660	5.098×10^{-3}
10	0.9950	1.197×10^{-1}
50	0.99980	1.426
100	0.99995	3.213

Table 17: Mid-bunch slice statistics, mesh-only. Worst relative error per component and fraction of unmasked plane pixels above 10^{-3}, x-y plane at $z = 1.60\,\mu$m and z-y plane at $x = 3.20\,\mu$m, $N = 1000$, $32 \times 32 \times 16$, cell 2×10^{-7} m, mesh against the independent exact TSC driver (FP64 quadrature, order 4, max level 5, tolerance 10^{-3}). The $\gamma = 10$ bunch moves along $+y$; y is the horizontal axis in every map. Node spacing $h = 1$ PIC cell; no near shell. The $\gamma = 1$ reference uses the archived tolerance 10^{-4} (driver floor 9.075×10^{-4}, 14 not converged).

plane	component	worst rel. err	frac $> 10^{-3}$
$\gamma = 10$			
x-y ($z = 1.60\,\mu$m)	E_x	1.81	54.1%
	E_y	1.16×10^{-1}	49.9%
	E_z	5.33	78.3%
	B_x	6.49×10^{-1}	72.7%
	B_z	3.77×10^{-1}	58.7%
z-y ($x = 3.20\,\mu$m)	E_x	5.06	90.4%
	E_y	1.17×10^{-1}	61.1%
	E_z	1.92	82.4%
	B_x	4.34×10^{-1}	80.9%
	B_z	7.16×10^{-1}	84.3%
$\gamma = 1$ *(stationary)*			
x-y ($z = 1.60\,\mu$m)	E_x	1.20×10^{-1}	22.8%
	E_y	1.34×10^{-1}	23.3%
	E_z	4.15×10^{-1}	74.7%
z-y ($x = 3.20\,\mu$m)	E_x	4.15×10^{-1}	49.6%
	E_y	1.37×10^{-1}	61.7%
	E_z	1.23×10^{-1}	53.9%

Table 18: Binned multi-velocity peaks, 64^3 grid, 200 sources, $\gamma = 10$.

spread	$B{=}16$ $\lvert B_z \rvert$ (T)	$B{=}64$ $\lvert B_z \rvert$ (T)	$B{=}16$ $\lvert B_y \rvert$ (T)	$B{=}64$ $\lvert B_y \rvert$ (T)
energy 5%	2.32050×10^4	2.32049×10^4	0	0
angular 5°	2.48202×10^4	2.48213×10^4	1.2948×10^3	1.2955×10^3

Table 19: Far-tail switch validation, 20-cell 4096-electron bunch, $R = 16.85$ cells, tolerance level 45.72 V/m (10^{-3} of the peak 45722.5 V/m). Worst per-component $|\mathbf{E}_{\text{tail}} - \mathbf{E}_{\text{direct}}|$ against an independent direct charge sum over 13 directions. The switch at $6R$ is the chosen switch: the error is 1.87×10^{-3} of the tolerance level.

switch radius	worst err (V/m)	/ tolerance level
$2R$ (33.7 cells)	4.80×10^1	1.05
$4R$ (67.4 cells)	8.77×10^{-1}	1.92×10^{-2}
$6R$ (101.1 cells)	8.53×10^{-2}	1.87×10^{-3}
$8R$ (134.8 cells)	1.66×10^{-2}	3.63×10^{-4}

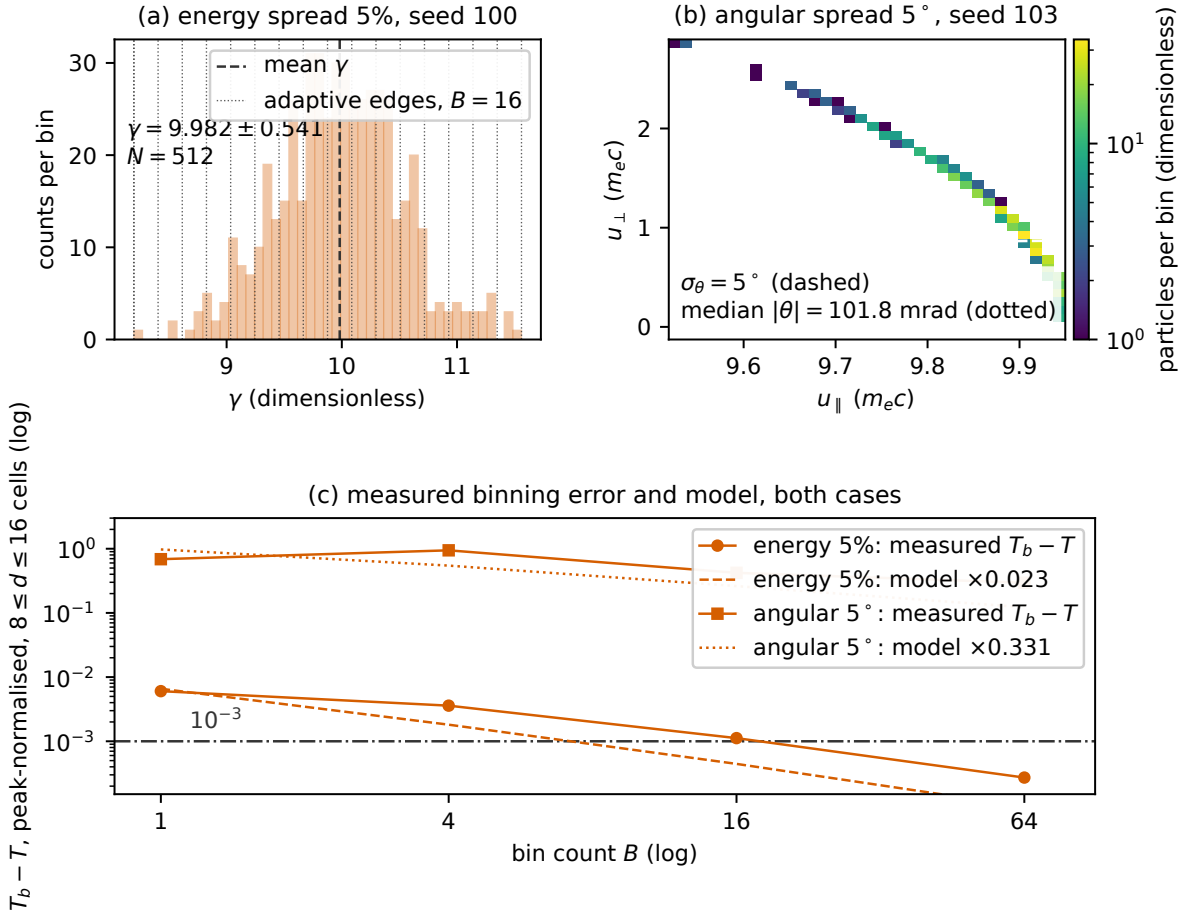


Figure 20: Sampled multi-velocity bunches ($\gamma = 10$, 512 sources, seeds 100 and 103); the $B = 16$ edges in (a) and the velocity boxes that produce the field come from the same `VelocityBinning` per-axis min/max path. (a) γ histogram for the 5% energy spread with the 16 adaptive edges drawn and the mean and standard deviation annotated; (b) $u_{\parallel}$–$u_{\perp}$ density for the 5° angular spread with the nominal 5° divergence and the measured median $|\theta| = 101.8$ mrad marked; (c) relative deviation of the $B = 16$ peaks from the $B = 64$ reference, 4.3×10^{-6} (energy B_z), 4.4×10^{-5} (angular B_z) and 5.6×10^{-4} (angular B_y), with $B_y \equiv 0$ for the energy case, so the field is bin-independent at the stated $2.32050 \times 10^4 / 2.32049 \times 10^4$ T and 5.22×10^{-2} ratio.